Who Pays, Who Adopts?
Efficiency and Equity of Residential Solar Policy

Dongchen He*

Latest Version Here

Abstract

This paper studies diverse residential solar subsidies within a nested discrete choice framework, introducing endogenous capacity choice and heterogeneous household preferences. Solar subsidies affect the intensive margin. Households install small solar panels when subsidies reimburse upfront investment costs, creating a high fiscal cost to achieve the capacity target. Furthermore, households respond heterogeneously to subsidies: switching from a subsidy based on future production to one that reduces upfront investment costs shifts solar photovoltaics adoption toward lower-income households. I propose a novel policy screening method that minimizes fiscal costs and maximizes welfare gain. The method of raising subsidies also shapes distributional outcomes. These findings highlight the importance of subsidy policy design.

Keywords: Photovoltaics, Renewable subsidy, Distributional effects, Policy design, Structural estimation. **JEL:** D12, D31, D63, Q52, Q58

---

*Dongchen He is at the School of Economics and Management, Tilburg University. d.he@tilburguniversity.edu. I appreciate the big support of my supervisors, Bert Willems and Ronald Huisman. I am grateful to my Ph.D. committee members Jan Boone, Natalia Fabra, Machiel Mulder, and Nicola Pavanini for their constructive suggestions. I am thankful to Stefan Ambec, Olivier De Groote, Reyer Gerlagh, Tobias Klein, Stefan Lamp, Mathias Reynaert, François Salanié, and Herman Vollebergh for their helpful feedback. I also thank participants at BSE Summer forum 2026 (Applied Industrial Organization, Barcelona), ASSA IAEE 2026 (Philadelphia), IAEE International 2025 (Paris) and EEA 2025 (Bordeaux), and seminars at Tilburg University and Toulouse School of Economics. I acknowledge Centraal Bureau voor de Statistiek for providing me with the Dutch administrative data. Finally, I thank CentER and the economics department at Tilburg University for financial support. The author declares that she has no relevant or material financial interests that relate to the research described in this paper. All errors are my own.

1 Introduction

Reducing carbon emissions from the energy sector is crucial for addressing climate change. Solar photovoltaic (PV) is the main renewable technology choice for the private sector and has expanded rapidly in recent years. By 2024, solar PV accounted for 42% of global renewable capacity. Compared with utility-scale PV, residential PV has several advantages: it does not require additional land, can be installed quickly, and directly involves households in the energy transition. However, residential PV is more expensive, and households tend to be more risk-averse than firms and do not fully account for the environmental externalities. As a result, many countries use generous subsidies to accelerate residential PV adoption. With strong policy support, residential solar accounts for about 17% of global solar capacity in 2024.[1]

Although all subsidies share the same decarbonization objective, they affect which households adopt and how much capacity the households would like to install. These endogenous consequences are not sufficiently studied but are important in two respects. First, subsidies change PV adoption behavior in ways that may carry substantial social and fiscal costs. Second, subsidies may disproportionately benefit higher-income households, resulting in regressive effects. This raises a key question for policymakers: Can a better design of residential PV subsidies address these unintended outcomes while achieving the goals of the energy transition?

To disentangle the effect of subsidy design on households' PV adoption decisions, I develop a nested discrete choice framework that models the decision based on the installation cost of solar panels, future revenues from adopting PV, electricity consumption, rooftop constraints, and, importantly, the structure of the subsidy. There are two key specifications. First, households decide not only whether to adopt but also the PV size once they adopt. Second, households across income quintiles have heterogeneous preferences regarding installation costs and future revenues. This structural framework is flexible for investigating different policy designs and circumvents the reduced-form econometric challenge of requiring randomized control and treatment groups for all policies.

The framework is applied to Dutch administrative data, with 2505906 households from 2020 to 2022.[2] The data provides information on household solar PV adoption status, installed capacity, and socio-demographic characteristics. The Netherlands operates a liberalized and mature electricity wholesale and retail market. By 2022, the Dutch residential PV adoption rate reached 26%, making it one of the highest

[1] International Energy Agency. https://www.iea.org/energy-system/renewables/solar-pv

[2] The raw data cover the entire population of Dutch households. Data processing leaves 2505906 households, which is still a large sample size.

in Europe and providing rich variation for estimation. The official subsidy in the Netherlands is called net metering, which offsets household electricity consumption from the public grid (henceforth, grid consumption) with electricity fed into the grid (henceforth, electricity feed-in) at the current retail price. Net metering was also widely used in 45 U.S. states at its peak, as well as in Canada and several European countries such as Belgium.

With the estimation results, I conduct counterfactual analyses by changing the policy design. In addition to net metering, I consider three other subsidies. The first is a lump-sum transfer paid to households upon adoption, regardless of PV size. Although less common in practice, lump-sum transfer provides a meaningful benchmark because they do not affect households' capacity choices. Then, I consider a proportional investment subsidy that reimburses part of the installation cost and is independent of future electricity production. This investment subsidy has been widely implemented, especially in the United States. The last subsidy is the feed-in tariff, which is popular in Europe and reimburses each kilowatt-hour (kWh) of electricity feed-in at a fixed price. Despite variations in implementation details, these four policies are broadly representative and capture the main mechanisms this paper aims to study. In the rest of the paper, the lump-sum transfer and proportional investment subsidy are referred to as *investment-based* subsidy, and the feed-in tariff and net metering are referred to as *production-based* subsidy. Beyond the single-policy design, I also investigate *policy combinations* that combine an investment-based subsidy and a production-based subsidy, and *policy screening* that enables households to choose their preferred policy from a policy menu.

The first finding is that the subsidy structure affects the intensive margin. Given the total PV capacity target, the results show that compared with lump-sum transfers, investment subsidies increase average PV capacity by 0.6 kilowatts (kW) per household adoption. Net metering creates a kink in the incentives and bunches capacity at the household consumption level, increasing the average capacity by 0.8 kW. Feed-in tariffs reward additional capacity with a high, constant price, resulting in capacity typically limited by rooftop area, and increasing PV capacity by 1.2 kW per adopted household. These intensive margins affect installation costs through economies of scale and grid costs based on the amount of feed-in. A smaller PV has a higher average installation cost while lowering the grid burden through higher self-consumption. The countervailing cost effects make the total social cost of different policies modest.

However, the intensive margin has important implications for the fiscal budget. In particular, investment-based subsidies lead to smaller PV size per adoption, requiring a higher subsidy level to achieve the capacity target through higher adoption rates. Increasing the extensive margin makes investment-based subsidies very costly, as both marginal and inframarginal households installing solar PV receive higher subsidies. As a result, I find that an investment subsidy can save only 10% of the fiscal costs of the feed-in tariff, even if households heavily discount future benefits. As a reference, De Groote and Verboven (2019) show that investment subsidies save 51% compared to feed-in tariffs, without accounting for the intensive margin.

The second finding is that households respond heterogeneously to different subsidies. Specifically, low-income households are much more sensitive to reductions in installation costs than to future revenue, whereas higher-income households are slightly more responsive to future revenue. These heterogeneous preferences imply that switching from a production-based subsidy to an investment-based subsidy shifts adoption toward lower-income households. Given the capacity target, switching from net metering to a lump-sum transfer increases the annual conditional adoption rate, defined as the number of new adoptions in the potential adoption market, by 2.7% in the bottom income quintile and only 0.2% in the top quintile. These heterogeneous preferences also suggest the potential for policy screening. Specifically, designing a policy menu that includes both a feed-in tariff and an investment subsidy, and allowing households to self-select the option that best fits their preferences, can reduce total fiscal expenditure by about 18% relative to the optimal single policy.

Last, I discuss the welfare redistribution of PV subsidies. I find that the design of the subsidy itself cannot avoid its regressive effects. Hence, it is important to consider how the subsidy is financed. I show that surcharges on electricity consumption lead to a relatively equal welfare change for all households, as long as net metering is avoided. An income tax is progressive regardless of the subsidy used. Regarding welfare levels, I find that screening yields greater welfare gains.

This paper comprehensively discusses residential solar policies within a unified structural framework of household decision-making by twisting the endogenous capacity choice and heterogeneous household preferences together. The closest paper by Bollinger et al. (2025) calibrates heterogeneous discount factors for low-income and high-income households, and also suggests reimbursing upfront costs to reduce regressivity. My paper differs by allowing households to choose installation capacity within the structural model, and explicitly capturing the intensive margins and associated welfare changes under diverse policy designs, which provide the countervailing cost effects that, to my best knowledge, have not been identified in previous studies.

There is a growing body of literature discussing the role of subsidies on solar PV installation. For instance, Burr (2016) uses a quasi-experiment in California and shows that the investment subsidy encourages more adoption, while the production subsidy is more efficient as it encourages adoption in optimal locations for solar electricity production. There are some other papers estimating the price elasticity of investment subsidies (Crago & Chernyakhovskiy, 2017; Gillingham & Tsvetanov, 2019; Hughes & Podolefsky, 2015). On the other hand, Aldy et al. (2023) study wind farm subsidies and give the opposite result. Comello and Reichelstein (2017) predict PV adoption in three cities in the US when a lower-than-retail tariff is paid to solar adopters and find that the adoption will not be affected as long as this tariff is above the levelized cost of electricity. With an input-output model, Eid et al. (2014) calculate the bills in different scenarios and net-metering designs, providing insights into the effect of net-metering policy on cost recovery and inequality. Londo et al. (2020) use the cash-flow model and investigate the effects of alternative policies on pay-back period, government cost, and amount of PV uptake by exogenously given parameters. Masciandaro et al. (2025) show how net metering affects adopters and non-adopters across different regions in the Netherlands. Böning et al. (2025) assess the effects of different incentive schemes with a reduced form estimation using regional data in Belgium. My paper contributes by using structural estimation with a large administrative dataset, and by conducting comprehensive and comparable counterfactual analyses within this framework.

This paper also contributes to the literature on the welfare effects of residential solar PV adoption, with a particular focus on its subsidy policies. Closely enough, Feger et al. (2022) investigate an optimal tariff design to incentivize residential solar PV adoption and avoid an enormous grid cost burden on non-adopters in Switzerland. They argue that consumption-based grid cost is less regressive than fixed grid cost because adopters are more affluent and less price sensitive to electricity price increases. Wolak (2018) uses distribution network price and installation from the three largest utilities in California and finds that residential solar capacity contributed two-thirds of increasing network prices from 2003 to 2016. Dauwalter and Harris (2023) further show that residential solar capacity has unequal environmental benefits, and there is no trade-off between efficiency and equity. Rather than documenting the unequal outcomes, this paper contributes by identifying the underlying mechanisms of inequalities and proposing a subsidy design that would improve this issue.

Last, this paper enriches the discussion on the inequity of environmental policies. For instance, Holland et al. (2019) examine the distributional effects of local air pollution from electric vehicle adoption in the US. Ito et al. (2023) demonstrate that price-elastic consumers are more likely to benefit from dynamic pricing. Känzig (2023) shows that the poor are more exposed to carbon pricing because they have a higher energy share and face a larger fall in income. This paper shows that residential solar subsidies have regressive effects similar to those of other environmental policies.

The rest of the paper is organized as follows. Section 2 describes the policy discussed in this paper. Section 3 specifies the structural model for residential solar PV adoption decisions. Section 4 describes the datasets and sample construction. Section 5 discusses the empirical results. Section 6 performs counterfactual analyses. Section 7 conducts several robustness checks. Section 8 concludes.

2 Residential Solar Benefits and Policies

In this section, I describe the economic benefits faced by households when deciding whether to adopt solar PV and how much capacity to install. I start with a situation without subsidy, after which I introduce the four policy instruments analyzed in this paper.

2.1 No subsidy case

Without any subsidy, households that adopt solar PV benefit in two ways. First, they save on their electricity bills by directly consuming part of their own electricity generation, referred to as *self-consumption*. Each kilowatt-hour consumed in this way offsets the need to purchase electricity from the grid and is therefore valued at the current electricity retail price. Second, households can sell any surplus electricity back to the grid, but only at the electricity wholesale market price, which is typically well below the retail price. The gap between retail and wholesale prices reflects the profit margin of energy companies as well as taxes and levies imposed by the government and system operators. See Appendix A for a brief background of Dutch electricity markets. Consequently, without subsidy, solar PV adoption is attractive when households consume a significant share of their own production, while exporting surplus to the grid is far less profitable.

2.2 Policy instruments

Household electricity consumption is inelastic, and electricity storage is limited, so solar PV adoption progress was slow. Subsidies are used to correct these market frictions. This paper focuses on four policy instruments: lump-sum transfers, investment subsidies, feed-in tariffs, and net metering. Each instrument increases the value of solar adoption, but they do so in different ways and with different implications for household capacity choices. See Figure 9 in Appendix B.

2.2.1 Lump-sum transfer

The first policy instrument is a lump-sum transfer, paid once a household adopts solar PV, regardless of the capacity of the PV system. This policy makes adoption itself more attractive because the household receives a reward upon installing solar PV. However, since the transfer does not depend on capacity, it does not distort how many panels households choose once they adopt. Its main role is to stimulate adoption at the extensive margin.

2.2.2 Investment subsidy

The second instrument is an investment subsidy. Here, the government reimburses a share of the upfront installation cost. There are some variations of this investment subsidy, such as a tax credit spread over the next few years or a zero-interest loan. By lowering the installation cost, this policy reduces the marginal cost of each kW of capacity. Households that would otherwise find solar panels too costly can now adopt, and those that would have adopted anyway are encouraged to install larger systems, especially when the solar PV installation exhibits economies of scale, which will be discussed in Section 4. In short, investment subsidies stimulate both the decision to adopt and the size of the installation.

2.2.3 Feed-in tariff

The third instrument is a feed-in tariff, where households are paid a guaranteed price far above electricity wholesale prices for each kWh of electricity they feed into the grid. This tariff provides a stable and proportional incentive to expand solar system size: every additional panel generates additional revenue at the fixed tariff. Therefore, it strongly affects the intensive margin, leading to larger capacity choices. This often results in a corner solution, where households install as many panels as their rooftop can accommodate.

2.2.4 Net metering

The fourth instrument is net metering, with which households can offset their electricity consumption with their own solar generation, receiving the retail price for kWh of electricity produced, up to a certain amount. Any surplus beyond that threshold is typically reimbursed at a lower price. This creates a kink in incentives: installing capacity up to the given amount is very attractive, but additional expansion beyond this point yields much less benefit. Thus, capacity choice would bunch at the threshold. An intuitive threshold is the level of household electricity demand.

2.2.5 Policy combination

It is common to combine multiple policy instruments. For instance, an investment-based subsidy can be implemented together with a production-based one such as a feed-in tariff or a net metering scheme, so that households receive both an upfront reduction in installation cost and revenue from electricity generation. Two production-based subsidies can also be available at the same time. For example, households were allowed to receive a feed-in tariff while also being able to net their electricity consumption in Belgium. In this paper, I only consider the policy combination of a feed-in tariff and an investment subsidy.[3]

2.2.6 Policy screening

Policy screening is defined such that the government provides a menu of policies, and the households can choose exactly one policy they prefer. I consider a screening policy in which households can choose between a feed-in tariff and an investment subsidy.

3 Model

3.1 Model setup

This section develops a micro-foundation for household decisions on solar PV adoption. I propose a static nested discrete choice model (McFadden, 1977) to describe this decision.[4] In each year t, a household denoted by i either chooses to install one of the nested solar panel capacities $j = 1, ..., 5$ or not adopt $j = 0$. Each j only differs in capacity sizes, and there are five types. $j = 1$ refers to capacity less than 2kW, $j = 2$ includes the capacity between 2 to 4kW, and for every subsequent category, each represents a 2kW increment. K_j is the median capacity level of each type j.

[3] I do not consider the combination with a lump-sum transfer because it is only a theoretical benchmark. I also do not consider combinations with the net-metering policy, because it is the *status quo* policy, and any additional subsidy would strictly increase adoption, which makes the results incomparable.

[4] De Groote and Verboven (2019) and Feger et al. (2022) use a dynamic model because solar modules were quite expensive before the 2010s and have declined over the years. Hence, the waiting value for households was high, even though some households could install solar panels in the current period rather than non-adoption, and a dynamic model was useful to capture this waiting incentive. However, this paper uses the data from 2019 to 2022, when module prices have significantly declined and stabilized, and the waiting value is relatively low. Furthermore, De Groote and Verboven (2019) find that the main estimates, such as the discount factor and the price sensitivity, do not change much in a static and dynamic setting.

Each household is subject to a physical rooftop constraint, such that the installed capacity cannot exceed the maximum that the roof can accommodate. Hence, not all alternatives j are available for some households.[5] I follow Feger et al. (2022) and assume that each square meter of house surface can accommodate 0.08 kW of capacity. All solar panel types are classified into *adoption* group, denoted by g, with nesting parameter σ capturing the correlation of utilities that consumers experience among different types in the group. As $\sigma \to 0$, the correlation between solar panel types disappears and the nested model reduces to a standard multinomial model. The nested structure relaxes the restriction of substitution between capacities to be the same as the substitution between adoption and non-adoption, which also reduces bias from unobserved correlations. Intuitively, PV adoption can be seen as a sequential decision process. Households first decide whether to adopt based on physical and financial feasibility. If they choose to adopt, they then determine a PV capacity considering the rooftop surface and electricity consumption. See Figure 1.

Each household receives an idiosyncratic taste for adoption ζ_{igt}, and a random taste shock ε_{ijt}. ζ_{igt} follows the unique distribution proposed by Cardell (1997) such that $\zeta_{igt} + (1-\sigma)\varepsilon_{ijt}$ is type I extreme value random variable.

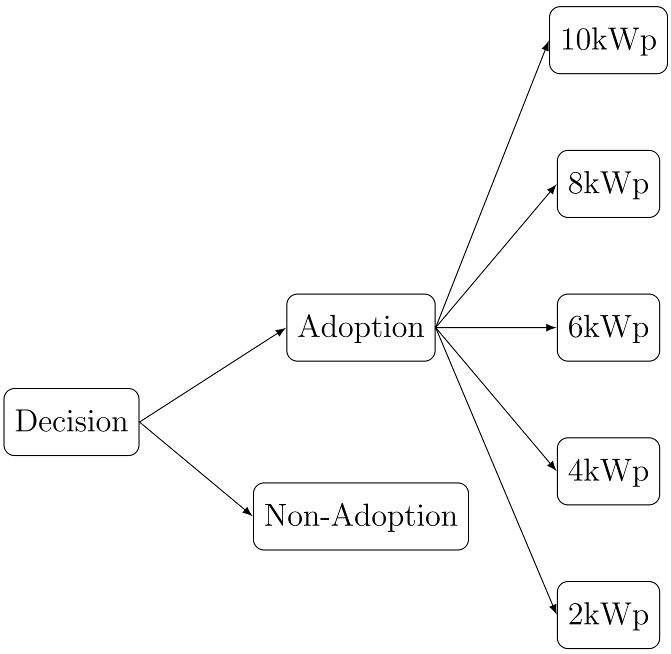


Figure 1: Nested Model Tree Diagram

[5]For notational simplicity, I do not introduce extra terms to define the choice set of each household i, but note that this is modeled in the estimation.

Value of adoption I describe the utility function of adoption as:

$$u_{ijt} = \delta_{ijt} + \zeta_{igt} + (1-\sigma)\epsilon_{ijt}, \tag{1}$$

where δ_{ijt} is the conditional value of household i adopting capacity type $j \in \{1,2,3,4,5\}$:

$$\delta_{ijt} = \beta^R_{q_i}\mathcal{R}_{ijt} - \beta^C_{q_i}\mathcal{C}_{ijt} + \mathbf{\Phi}_j\mathbf{w}_i + \boldsymbol{x_{ijt}\gamma}. \tag{2}$$

$\mathcal{R}_{ijt}$ denotes the future revenue (including production-based subsidy) from installing solar PV, and $\mathcal{C}_{ijt}$ is the cost of installation (net of investment-based subsidy). $\beta^C_{q_i}$ measures the price sensitivity to the upfront installation cost, while $\beta^R_{q_i}$ captures the sensitivity to the future revenue. I assume that household income is divided into five quintiles, $q_i \in \{1,2,3,4,5\}$, and price sensitivity is allowed to differ across these quintiles. I also allow $\beta^C_{q_i} \neq \beta^R_{q_i}$ to capture the liquidity constraints and the discounting difference across income quintiles.[6] $\mathbf{w}_i$ is a K-dimensional vector of household characteristics, consisting of housing type, year of construction, age of the household's reference person, household wealth and income, as well as house size and household size. $\mathbf{\Phi}_j$ is a $K \times 1$ vector of parameters. For house size and household size, I assume that $\phi_j = \phi \cdot j$, while for the remaining household characteristics, I assume that ϕ_j is constant across all $j \neq 0$. $\boldsymbol{x_{ijt}}$ denotes a vector of control variables, including alternative-location fixed effects, and year fixed effects. The parameters of interest are $\Theta = \{\boldsymbol{\beta^C_i}, \boldsymbol{\beta^R_i}, \sigma, \boldsymbol{\gamma}, \boldsymbol{\Phi_j}\}$.

Value of non-adoption The utility function of non-adoption is:

$$u_{i0t} = \delta_{i0t} + \epsilon_{i0t}. \tag{3}$$

Without loss of generality, δ_{i0t} is normalized to be 0.

Estimating strategy For each household i, the probability of choosing each type $j = 0, 1, ..., 5$ takes a well-known closed-form expression,

[6]The available data do not allow me to separately identify liquidity constraints and heterogeneity in discounting in a clean way. The reason is that the discount factor used to compute discounted future revenue rescales the parameter $\beta^R_{q_i}$, so the discount factor and $\beta^R_{q_i}$ cannot be identified simultaneously. I therefore need to choose between two model specifications: either estimate the discount factor while imposing $\beta^C_{q_i} = \beta^R_{q_i}$, or estimate both $\beta^C_{q_i}$ and $\beta^R_{q_i}$ while treating the discount factor as exogenously given. In the main specification, I choose the latter specification as all parameters enter the conditional utility linearly, which substantially reduces the computational burden. Section 7 shows that changing the specification does not alter the main results of the paper.

$$s_{ijt} = \frac{1}{1+\exp(I_{igt})}, \quad j = 0, \tag{4}$$

$$s_{ijt} = \underbrace{\frac{\exp\left(\frac{\delta_{ijt}}{1-\sigma}\right)}{\sum_{l=1}^{5}\exp\left(\frac{\delta_{ilt}}{1-\sigma}\right)}}_{s_{j|g,t}} \cdot \underbrace{\frac{\exp(I_{igt})}{1+\exp(I_{igt})}}_{s_{gt}}, \quad j = 1,\ldots,5, \tag{5}$$

where I_{igt} is the inclusive value of adopting solar panels:

$$I_{igt} = (1-\sigma)\cdot\log\left(\sum_{l=1}^{5}\exp\left(\frac{\delta_{ilt}}{1-\sigma}\right)\right). \tag{6}$$

Let $y_{ijt} = 1$ if j is chosen by household i in year t, and otherwise $y_{ijt} = 0$. The sum of the log likelihood of the adoption decision of all households across all years is calculated as:

$$\mathcal{L}(\Theta) = \sum_{t}\sum_{i=1}^{N}\sum_{j=0}^{5} y_{ijt}\,\log s_{ijt}(\Theta). \tag{7}$$

Hence, the parameters can be estimated by maximizing the sum of the log-likelihoods.

$$\hat{\Theta} = \arg\max_{\Theta}\ \mathcal{L}(\Theta). \tag{8}$$

3.2 Calculation of $\mathcal{R}_{ijt}$ and $\mathcal{C}_{ijt}$

In this part, I describe how the future revenue and the net installation cost of adopting solar PV are calculated. I start with the net installation cost $\mathcal{C}_{ijt}$, which is defined as the gross PV installation cost minus the lump-sum transfer and the investment subsidy:

$$\mathcal{C}_{ijt} = \underbrace{C_{ijt}(K_j)}_{\text{gross cost}} - \underbrace{T\cdot\mathbf{1}\{K_j>0\}}_{\text{lump-sum}} - \underbrace{\kappa C_{ijt}(K_j)}_{\text{investment subsidy}}. \tag{9}$$

C_{ijt} is the PV installation cost that benefits from economies of scale, $C_{ijt}(0) = 0, \frac{\partial C}{\partial K} > 0, \frac{\partial^2 C}{\partial K^2} \leq 0$. T is the lump-sum transfer that is paid if a household adopts solar PV, no matter what the capacity size is, and κ is the proportion of investment subsidy that reimburses part of the gross installation cost. $T = 0$ indicates that there is no lump-sum transfer, and $\kappa = 0$ indicates that there is no investment subsidy.

I then describe the calculation of discounted future revenue $\mathcal{R}_{ijt}$. As explained in Section 2, there are two parts of revenue. The first is from self-consumption. Then, households can also sell excess electricity at the market price or an agreed price, depending on which production-based subsidy is used.

$$\mathcal{R}_{ijt} = \mathcal{R}_{ijt}^{SC} + \mathcal{R}_{ijt}^{Feed-in}, \tag{10}$$

where

$$\mathcal{R}_{ijt}^{Feed-in} = \begin{cases} \mathcal{R}_{ijt}^{M} & \text{no subsidy: receive market value} \\ \mathcal{R}_{ijt}^{FIT} & \text{feed-in tariff} \\ \mathcal{R}_{ijt}^{Net} & \text{net metering} \\ \mathcal{R}_{ijt}^{FIT} + \mathcal{R}_{ijt}^{Net} & \text{policy combination} \end{cases} \tag{11}$$

Table 1 summarizes five parameters and six variables used to calculate the revenue $\mathcal{R}_{ijt}$. For parameters, there are the discount factor ρ and the inflation rate π. There are three other parameters specific to solar PV. Solar efficiency ι measures how effectively the panel converts solar energy into electricity, expressed as the annual electricity production divided by capacity, kWh/W. The self-consumption rate α is defined as self-consumption over total solar electricity production. λ is the solar panel's depreciation factor.

The variables and parameters could vary with household, PV capacity, or time, indicated by a subscript. For instance, Y_j is production from solar PV of type j.

Table 1: Summary of Parameters and Variables

Parameter	**Definition**	**Variable**	**Definition**
ι	Solar efficiency	D	Electricity demand
α	Self-consumption rate	Y	Electricity production
ρ	Discount factor	Z	Electricity feed-in $= (1-\alpha)Y$
λ	Depreciation factor	R	Electricity retail price
π	Inflation rate	p_c	Feed-in tariff
		p_s	Electricity wholesale price/reduced rate

Define ρ' as the adjusted discount factor:

$$\hat{\rho} = \rho/(1+\pi). \tag{12}$$

Assume the life span of a solar panel is 25 years. The revenue functions are:

1. Revenue from self-consumption:

$$\mathcal{R}_{ijt}^{SC} = \sum_{k=t}^{t+24} \hat{\rho}^{k-t}\mathbb{E}_t[R_k] \cdot \mathbb{E}_t[Y_{ik} - Z_{ik}]. \tag{13}$$

2. Revenue from the wholesale market:

$$\mathcal{R}_{ijt}^{M} = \sum_{k=t}^{t+24} \hat{\rho}^{k-t}\mathbb{E}_t^k[p_{sk}] \cdot \mathbb{E}_t[Z_{jk}]. \tag{14}$$

3. Revenue from feed-in tariff:

$$\mathcal{R}_{ijt}^{FIT} = \sum_{k=t}^{t+24} \hat{\rho}^{k-t} p_c \cdot \mathbb{E}_t[Z_{jk}]. \tag{15}$$

4. Revenue from net metering:

$$\mathcal{R}_{ijt}^{Net} = \sum_{k=t}^{t+24} \hat{\rho}^{k-t} \{\mathbb{E}_t[R_k] \cdot \min\{\mathbb{E}_t[Z_{ik}], \mathbb{E}_t[D_{ik}] - \alpha_{ij}\mathbb{E}_t[Y_{ik}]\} + \mathbb{E}_t[p_{sk}] \cdot \max\{0, \mathbb{E}_t[Y_{ik}] - \mathbb{E}_t[D_{ik}]\}\}. \tag{16}$$

I assume that households form a naive expectation of future electricity prices and their electricity consumption. Specifically, households make adoption decisions based on one-year lagged electricity consumption, which is assumed to remain relatively stable over time. Hence, $\mathbb{E}_t[D_{ik}] = D_{i,t-1}$.[7] Also, retail and wholesale prices are fixed at the levels in the year of adoption and remain constant over the lifetime of the solar panels. $\mathbb{E}_t[R_k] = R_t$ and $\mathbb{E}_t[p_{sk}] = p_{st}$. Electricity production expection follows the function: $\mathbb{E}_t[Y_{jk}] = (1-\lambda)^{k-t}\iota K_j$.

4 Data

I estimate the model in Section 3 in the context of the Dutch residential solar PV market. In the Netherlands, the official subsidy is net metering, which was formally introduced in 2004.[8] Under this scheme, households with solar panels can offset their

[7]Note that only electricity consumption from the grid and electricity fed back into the grid are recorded, while self-consumption is unobserved. Once a household installs solar panels during the year, its total electricity consumption can no longer be directly observed. Therefore, I use one-year lagged consumption as a proxy. This assumption implies that households' electricity consumption remains unchanged regardless of solar PV adoption. Aydın et al. (2023) find the rebound effect in the Netherlands is only 0.07, and the data shows that regressing residential solar capacity on lagged electricity consumption gives a coefficient not significantly different from one, implying that during the sample period, households do not consider the long-term consumption change. Energy efficiency improvements and the adoption of heat pumps or electric vehicles have received increasing attention in recent years and may affect household electricity consumption. However, modeling these dynamics is beyond the scope of this paper, and I leave it as an interesting topic for future research.

[8]Before 2004, there was also "unofficial" net metering because the discs on traditional meters would spin backward when electricity was fed into the grid. Between 2008 and 2010, residential solar PV also benefited from feed-in premiums on top of net metering. After 2011, net metering became the sole incentive policy. In addition, low-interest loans under the program "National Heat Fund" (Warmtefonds) are available for low-income households, but their conditions have changed over time. Since these loans are not a default subsidy and applications are unobservable, they are not included in the analysis.

grid electricity consumption with electricity feed-in. Therefore, they save on the retail energy price and avoid paying energy tax and value-added taxes (VAT) on electricity. Grid costs, however, are fixed and unaffected by net metering. Initially, households could net a maximum of 3000 kWh per year, which was increased to 5000 kWh in 2011. Since 2013, the upper limit has been abolished, and households can offset up to 100% of their annual electricity consumption from grid.[9]

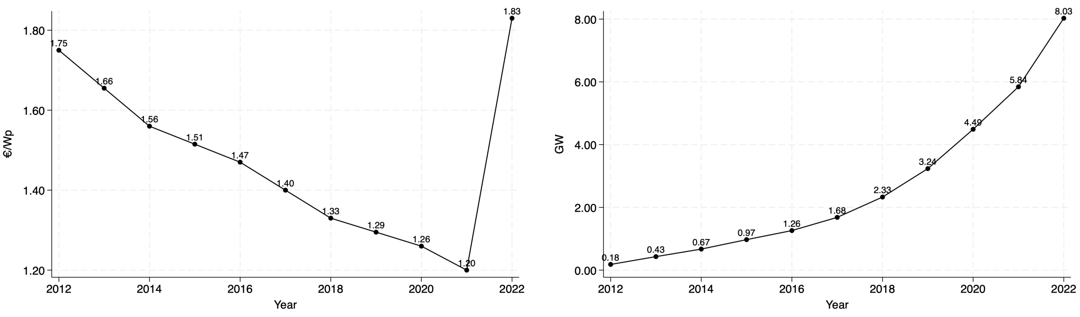
Several datasets are used for this research. First, I obtained detailed household data from the Centraal Bureau voor de Statistiek of the Netherlands (henceforth: CBS). These data include household yearly electricity consumption from the grid and electricity feed-in to the grid, as well as solar PV adoption status and adopted capacity from 2019 to 2022. They also offer socio-demographic attributes such as household wealth and income, and dwelling characteristics such as residence type and surface areas. I also have information on solar PV installation costs, electricity wholesale and retail prices. These data are used for estimating the nested discrete choice model developed in Section 3. The second key dataset is the average household grid consumption and feed-in profile from 2020 to 2022 at 15-minute frequency from MFFBAS. I use these data to estimate the self-consumption rate and the solar PV efficiency rate, which are exogenous inputs to the discrete choice model.

PV costs and adoption Residential solar PV installation costs consist of solar PV module costs, inverter costs, labor costs, and other material and operating costs. Until 2023, households paid VAT of 21% when purchasing solar panels, but this tax could be fully reclaimed. Since 2023, there has been no VAT on solar panels unless someone gets roof-integrated PV panels when buying a newly built house. In this case, solar panels are considered part of the roof and need to be paid VAT. The PV installation costs vary according to the type of PV modules and the roof. For instance, Monocrystalline cells have higher efficiency and are 20-30% more expensive than standard polycrystalline cells. Installing solar panels on a sloping roof is more costly than on a flatter one. Furthermore, as module costs have dramatically decreased over the past few years, labor costs take a larger share in the breakdown of installation costs. Since labor and operating costs do not increase linearly with PV capacity, residential PV installations benefit from economies of scale, implying a decreasing marginal cost. I calibrate that, compared to the smallest capacity size, the capacity

[9] As of June 2024, energy companies are allowed to charge a feed-in fee to PV households for returning electricity to the grid. Furthermore, the Dutch government has announced the complete phase-out of the net metering policy, effective January 1, 2027. From that date, PV adopters will no longer be able to offset their grid consumption with feed-in. Instead, they will pay retail prices for grid consumption and receive a “reasonable compensation” price for electricity fed into the grid.

2-4 kW offers a 40% reduction in unit price, to increase capacity to 4-6 kW offers a 12% reduction in unit price, and the further upgrade of one capacity type leads to a 5% reduction in unit price.[10]

Figure 2a depicts the simple average price per watt-peak (Wp) from 2012-2022.[11] There was a price spike in 2022 because soaring electricity prices led to a substantial surge in installations, while supply was low due to the COVID-19 pandemic.[12] The Netherlands has witnessed a steady growth in residential solar PV adoption. See Figure 2b. 26.23% of Dutch households had residential PV by 2022.



(a) Installation cost (€/Wp) **(b)** Accumulated capacity

Figure 2: Residential solar PV in the Netherlands.

Notes: Panel (a) shows the average price (VAT excluded) of residential solar installation per watt-peak, sourced from Milieu Centraal. Panel (b) shows the total installed residential solar capacity, accumulated at the end of each year, published by CBS.

Grid consumption and feed-in profiles The grid consumption profile measures how much electricity a household draws from the grid over time, and the feed-in profile refers to how much surplus electricity from residential solar is fed back into the grid. MFFBAS collects power flow data every fifteen minutes, and then sums the

[10]The installation costs differ among installation companies, the location, and the specific installation time, while the data are not available. Retail electricity prices also depend on the energy company and the month when the energy contract is signed. However, as the solar panel market and the electricity retail market are competitive, the cross-sectional price difference should not raise a big issue.

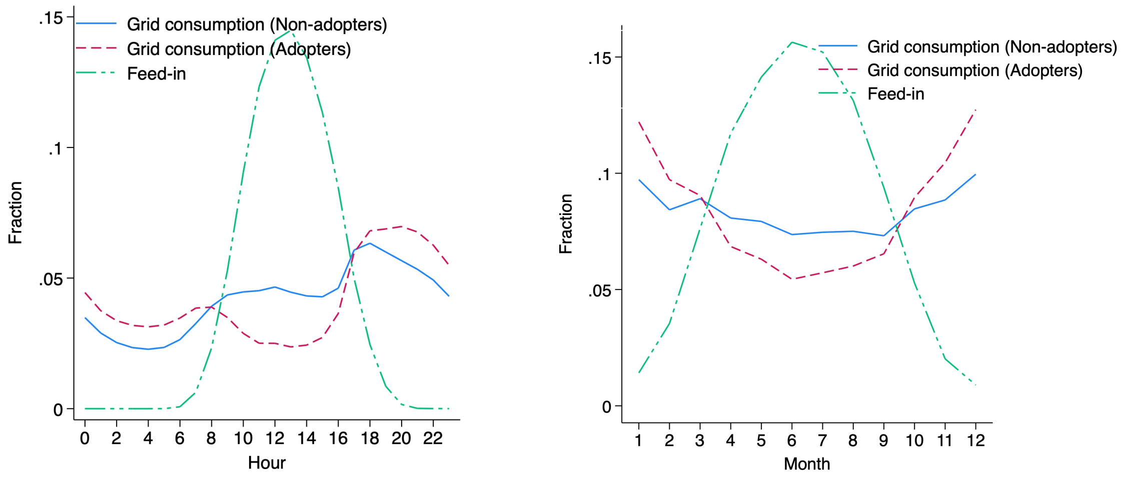
[11]Watt-peak measures the maximum power of a solar panel. For instance, one kWp can generate 1 kWh on an ideally sunny hour. However, depending on local weather conditions and solar panel deterioration, one kWp generates less than 1 kWh in one hour.

[12]Milieu Centraal did the market research once every two years and made a linear regression based on all the data collected in the previous years. After consulting with Milieu Centraal, the mean price between survey years is a good approximation of skipped years. The prices in the years 2013, 2015, 2017, and 2019 are estimated data.

flow data and publishes the *fraction* of flow over the annual total. Hence, profiles are normalized grid consumption and feed-in so that the sum of fraction equals one. Separate profiles are available for households with and without solar panels.

Figure 3 aggregates the profiles into hourly and monthly levels. For instance, the feed-in fraction from 1 to 2 p.m. is 0.145, meaning this one hour contributes to 14.5% of the annual feed-in. Similarly, the feed-in fraction in June is 0.156, meaning this month accounts for 15.6% of the total feed-in. The feed-in profile exhibits evident daily and seasonal patterns, peaking in the afternoon and summer.

Grid consumption is higher in the evening and winter. A typical daily consumption profile has two peaks. One is around 8:00 AM, when people wake up and start to work, while solar electricity takes a small share of total production. The other one is at 7:00 PM. After sunset, the demand for lighting increases. Furthermore, grid consumption profiles differ between PV adopters and non-adopters, because PV adopters can directly use part of the electricity produced by solar panels, resulting in a smaller fraction during the afternoon and summer. This difference in grid consumption allows me to estimate the self-consumption rate and solar efficiency, as shown in Appendix C.



(a) Hourly Profile **(b)** Monthly Profile

Figure 3: Grid Consumption and Feed-in Profile

Notes: This figure shows the average electricity grid consumption and feed-in profiles for Dutch households from 2020 to 2022. Quarter-hourly data are collected from MFFBAS and summed up at monthly and hourly levels.

Household data The full sample from CBS comprises Dutch population data from 2019 to 2022. To accurately merge the household demographic attributes and energy data, I made several sample restrictions. First, I dropped the households that moved within a year. Second, I dropped the households with unverified electricity consumption. Furthermore, I dropped the households that shared the same dwelling since assigning dwelling-based electricity grid consumption and feed-in to each household was impossible. Finally, households with PV capacity larger than 10kW were removed to rule out commercial PV systems.

Table 2 gives descriptive statistics of cleaned data for the snapshot from 2022. In total, there are 5446262 households, and 1491062 have adopted solar panels. We observe differences between PV adopters and non-adopters. All households are classified into five income quintiles based on the raw data: the first quintile represents the bottom 20% of households (poorest), while the fifth quintile represents the top 20% (richest). PV adoption varies significantly across different income quintiles. The average adoption rate is 27%,[13] from 12% for the poorest households and 44% for the richest ones. There are several potential reasons to explain the distinct adoption rates. First, high-income households have fewer financial constraints and can bear high installation costs. Second, homeowners are more likely to install residential solar than tenants. 93% of the richest households are homeowners, while only 9% of households of the first income quintile live in their own house. We also observe that poorer households tend to reside in apartments and do not have their own independent roofs. Finally, there are clear capacity differences among households that have adopted solar panels, and the installed capacity is close to electricity consumption.[14]

Sample construction To precisely identify PV adoption behavior, additional steps are taken to construct the final estimation sample. First, I only keep households that

[13]The adoption rate across the Netherlands is 26.23%. Hence, the sample used in this paper is representative.

[14]One may notice that PV capacity is larger than grid consumption. This is because part of the electricity demand is met directly by solar generation (self-consumption), which is not recorded by the meter. This is referred to as behind-the-meter consumption.

Table 2: Summary Statistics

Panel A: PV adopters vs non-adopters

	Non-adopter	Adopter
Age	57.47	56.37
HH size	2.01	2.57
Ownership	0.55	0.79
House	0.59	0.94
Wealth percentile	50.00	62.41
Income percentile	47.50	63.63
House size (m^2)	108.82	140.88
Grid consumption (kWh)	2547.37	2821.91
# of Obs	3955200	1491062

Panel B: attributes by income quintile

	# of Obs	All	< 20%	20–40%	40–60%	60–80%	> 80%
Dispo-income (€)	5446262	50859	19364	29607	42152	60110	100591
Grid consumption (kWh)	5446262	2623	1694	2048	2455	3032	3778
Feed-in (kWh)	5446262	523	152	237	458	730	994
House size (m^2)	5446262	118	83	99	116	130	156
Ownership (%)	5446262	62	9	40	72	84	93
House (%)	5446262	68	37	56	71	83	89
Adoption (%)	5446262	27	12	16	26	36	44
PV Capacity (kW)	1491062	3.62	2.40	2.77	3.33	3.79	4.27
Feed-in (kWh)	1491062	1910	1229	1469	1788	2005	2239

Notes: Panel A compares household characteristics between PV adopters and non-adopters in 2022. Panel B describes the average PV adoption rate, adopted capacity, disposable income, house ownership, electricity grid consumption, and feed-in for five income quintiles of Dutch households in 2022.

have full observations from 2019 to 2022.[15] Second, households installed PV in year

[15] 53.6% of observations belong to households observed in all four years, 14.7% have at least one skipped year in between, and 31.7% correspond to households that move during the sample period, or these households may simply have missing observations in a given year, even though they are observed in consecutive years before and after. Skipping years creates an identification problem for installations. For example, if PV is observed in year t but the data in year $t-1$ is missing, it is not possible to determine whether the installation occurred in year t or $t-1$. I therefore remove households with skipped years. Similarly, I drop households that move, as lagged housing and household characteristics are not observed and cannot be used for installation estimation. The average adoption rate over the sample period is 19.7% for households observed in all four years, 19.9%

t are removed from the potential market in year $t+1$, since solar PV adoption is a terminal action.[16] Third, identifying PV adoption in 2019 is impossible, as I cannot distinguish whether an adopter installed a solar system in 2019 or before 2019. Therefore, only data from 2020 to 2022 remains, giving an unbalanced panel covering 2505906 households over three years. The welfare analysis will be based on this sample (welfare sample).

I only keep homeowners for adoption estimation. The welfare sample shows an annual average adoption rate of 2.52% among tenants. However, the adoption difference is only 1.64% between the first and fifth quintile, while the difference among homeowners is 8.5%. Hence, it is difficult to argue whether the decision is made by agencies, landlords, or tenants. The estimation is conducted on a 10% random subsample of the data. Each household has up to six alternatives, and the resulting estimation sample comprises 2314295 observations (estimation sample).

5 Results

The *status quo* policy in the Netherlands is net metering policy that reimburses all surplus solar electricity up to the level of electricity consumption from the public grid. Hence, the total discounted *status quo* revenue is:

$$\mathcal{R}_{ijt}^{sq} = \sum_{k=t}^{t+24} \hat{\rho}^{k-t} \left(R_t \min\{D_{i,t-1}, Y_{jk}\} + p_{st} \max\{Y_{jk} - D_{i,t-1}, 0\} \right). \tag{17}$$

and the *status quo* cost is:

$$\mathcal{C}_{ijt}^{sq} = C_{ijt}. \tag{18}$$

5.1 Estimation outside of nested logit model

Parameters in Table 1 are entered exogenously into the nested logit model. I follow De Groote and Verboven (2019) and set the yearly discount factor $\rho = 0.85$. $\pi = 3\%$ is the inflation rate borrowed from a wide range of literature and close to the actual average inflation rate in the Netherlands.[17] The depreciation factor λ is set to be 3%

for households with missing years, and 20.3% for moving households, suggesting that dropping these observations is unlikely to introduce systematic bias.

[16] PV renovation or removal is very rare in the data. I observe fewer than 100 cases in this large dataset.

[17] The average yearly inflation rate in the Netherlands from 2012 to 2022 is 2.47%. The yearly inflation rates from 2019 to 2022 are 2.67%, 1.11%, 2.82% and 11.62% respectively. The low inflation in 2020 caused by COVID-19 and the exceptionally high inflation in 2022 driven by the energy crisis are unsustainable and unrepresentative.

in the first year and 0.7% afterward, according to Feger et al. (2022).

I use the grid consumption and feed-in profiles to estimate the average self-consumption rate α and the efficiency rate ι. Appendix C gives $\alpha = 0.33$ under the current net metering policy, suggesting that households, on average, directly use 33% of electricity produced by residential solar systems.[18] Furthermore, the efficiency rate is estimated to be $\iota = 0.91$.[19] Hence, in the Netherlands, 1kW of PV capacity can, on average, produce 910kWh per year.

5.2 Estimation of nested logit model

This section reports the parameter estimates in equation (8). Table 3 consists of two sets of results. The column (1) gives the standard model estimates, and the column (2) presents the PV adoption elasticity. The estimates show a large alternative correlation parameter (σ), validating the nesting structure of the model. As expected, installation costs negatively affect adoption decisions, while future revenue has a positive impact. The parameter β to future revenue is higher than that of the installation cost.[20] The terms $\Delta\beta$ capture the interaction of income quintiles with installation costs or future revenue. The results show that poorer households are more price sensitive than richer households. More importantly, the difference in price coefficients between costs and revenue is greater for lower-income households, $\Delta\beta^C_{q1} - \Delta\beta^R_{q1} > \Delta\beta^C_{q2} - \Delta\beta^R_{q2} > ... > \Delta\beta^C_{q4} - \Delta\beta^R_{q4} > 0$. Hence, reallocating an equivalent subsidy from future revenue to cost reduction yields a larger utility improvement for poorer households. Moreover, note that the ratio β^C/β^R changes with income quintile monotonically, implying the feasibility of tailoring policy instruments to different income groups. I will discuss this in more detail in Section 6.

The results in column (2) show that the demand for residential PV is very elastic. For instance, a 1% increase in discounted future revenue would increase the adoption probability for first quintile households by 7.147% and for fifth quintile households by 5.268%, while a 1% decrease in installation cost would increase the adoption probability for first quintile households by 10.513% and for fifth quintile households by 4.155%. Poorer households are more sensitive to installation costs than to future

[18]I further obtain that the amount of self-consumption accounts for 31% of the household's total electricity consumption. This also means that under the current net metering policy, households install capacity that is very close to their actual electricity consumption.

[19]The solar efficiency typically ranges from 0.85 to 0.95, depending on weather conditions and module efficiency.

[20]The estimate of β^R changes with the assumption of discounting factor ρ. I will discuss this in Section 7.

revenue, which may reflect liquidity constraints.[21] Interestingly, households in the fifth income quintile (the richest group) exhibit higher elasticity to future revenue. One explanation is that rich households adjust the size of the PV system rather than their adoption decision when costs change. Also, rich households adopt larger PV systems and benefit from economies of scale, so future revenue changes are more salient than cost changes.

The estimates on other household characteristics are intuitive. For example, households living in apartments face a high fixed cost when installing solar panels, which can be interpreted as an information friction or an installation obstacle in the absence of an independent roof. The cost of this friction is substantial. For example, in specification (4), the coefficient of -2.396 on apartment friction corresponds to an additional cost of €5023 for households in the first income quintile and €13385 for those in the highest quintile. When the household head is older, adoption is less likely, although this effect is very small. Moreover, newly built dwellings are more likely to install solar panels. Higher-income households are also more inclined to adopt, but this pattern does not exist across wealth quintiles.[22] With respect to solar PV size, households living in larger dwellings or with larger household sizes tend to install larger PV systems. Finally, the year 2022, with exceptionally high electricity prices, is associated with a surge in solar PV installations.

[21] Green loans are available to low-income homeowners. However, the data show that only 1% of PV adoptions in 2022 used this loan. This may reflect market failures such as asymmetric information or behavioral friction, such as limited attention or hassle costs. Also, I impose the same discount factor for all households, there is an alternative specification could be that households have heterogeneous intertemporal preferences. See Section 7.

[22] Table 2 shows that PV adopters hold both more wealth and income. However, once home ownership is controlled, wealth differences vanish, while income differences remain.

Table 3: Estimation Results

	(1) Estimates	(2) Elasticity
Revenue sensitivity(β^R, €10^3)	0.304*** (0.014)	5.268
Cost sensitivity ($-\beta^C$, €10^3)	-0.179*** (0.011)	4.155
Interactions with revenue		
first income quintile ($\Delta\beta^R_{q_1}$, €10^3)	0.219*** (0.043)	+1.878
second income quintile ($\Delta\beta^R_{q_2}$, €10^3)	0.217*** (0.020)	+2.026
third income quintile ($\Delta\beta^R_{q_3}$, €10^3)	0.112*** (0.011)	+0.876
fourth income quintile ($\Delta\beta^R_{q_4}$, €10^3)	0.042*** (0.008)	+0.288
Interactions with cost		
first income quintile ($-\Delta\beta^C_{q_1}$, €10^3)	-0.298*** (0.037)	+6.358
second income quintile ($-\Delta\beta^C_{q_2}$, €10^3)	-0.297*** (0.016)	+6.638
third income quintile ($-\Delta\beta^C_{q_3}$, €10^3)	-0.145*** (0.009)	+3.202
fourth income quintile ($-\Delta\beta^C_{q_4}$, €10^3)	-0.053*** (0.008)	+1.135
Apartment	-2.396*** (0.047)	
HH age	-0.003*** (0.001)	
House age	0.089*** (0.003)	
Wealth (€10^5)	-0.013*** (0.001)	
Income (€10^4)	0.008*** (0.002)	
House size ($100m^2$)	0.061*** (0.003)	
HH size	0.008*** (0.002)	
Year 2022	0.737*** (0.020)	
σ	0.764*** (0.009)	
# of Obs	2,314,295	1,890,020

Notes: The column (1) reports estimation results for PV adoption from 2020 to 2022. Standard errors in parentheses are clustered at the household level. The column (2) reports the average own-price elasticity across the five PV capacity types. The first two rows show the elasticity for the fifth income quintile. For all subsequent rows, the values represent the change in elasticity relative to the fifth quintile for the corresponding income group. Fixed effects included but not shown. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

6 Counterfactual

This section performs a counterfactual analysis based on the estimation results in Section 5. I first examine how different subsidy instruments affect households' decisions to adopt solar PV, focusing on how policy design shapes the adoption distribution across income groups and the choice of PV size. I then discuss the cost implications of these endogenous effects. Finally, I assess the welfare effects of the alternative subsidies.

6.1 Intensive margin and adoption distribution

To begin, it is crucial to realize that the government not only decides which subsidy to use, but also the level of subsidy. To make meaningful comparisons, it is necessary to establish a benchmark to calculate the subsidy level. I set a capacity target as the *status quo* policy net metering. This is a practical and reasonable target as it directly links to the climate goal of reducing a certain amount of carbon emissions. Then, I calculate the subsidy rate for every single policy to achieve the same PV capacity. The policy combination and screening concerns two subsidy variables and can only be determined with one more restriction, which I will discuss in Section 6.2. The adoption capacity is calculated by the equation below.

$$\text{Total capacity} = \text{Market size} \times \text{Conditional adoption rate} \times \text{Capacity per adoption} \tag{19}$$

The market size is the number of households that would potentially adopt PV, and the conditional adoption rate is defined as the share of adopting households over the market size.[23] The market size is an exogenous number that does not depend on policy design. The conditional adoption rate captures the extensive margin, while the capacity per adoption reflects the intensive margin. If a policy does not affect the intensive margin, the extensive margin should also be the same across all policies to achieve the same total capacity target. Otherwise, a large intensive margin implies a smaller extensive margin, and vice versa.

Table 4 reports the counterfactual outcomes. Panel A presents the conditional adoption rate and its distribution across the five income quintiles defined by the raw data. Two patterns emerge. First, investment-based subsidies give higher adoption rates than production-based subsidies. For example, the adoption rate under a lump-sum transfer is 2.09% higher than under a feed-in tariff. Second, the adoption rate of low-income households increases more than that of high-income households when switching from a production-based subsidy to an investment-based one. This pattern is aligned with the estimation results, which show that low-income households are

[23]This is distinguished from the unconditional adoption rate, which is defined as adoption over the total population.

more sensitive to reduction of upfront installation costs than to future revenue, and that this cost–income sensitivity gap is larger for low-income households than for high-income households.

Table 4: Adoption Distribution of Alternative Policies

	All	$< 20\%$	20–40%	40–60%	60–80%	$> 80\%$
Panel A: conditional adoption rate (%)						
No subsidy (baseline)	3.62	1.49	1.57	2.94	4.98	6.78
Net metering (*status quo*)	6.70	2.68	2.89	5.06	8.12	10.53
Feed-in tariff	6.04	2.76	2.94	4.74	7.20	9.14
Investment subsidy	7.14	3.95	4.25	5.80	8.08	9.87
Lump-sum	8.13	5.38	5.72	6.97	9.10	10.70
Panel B: capacity per adoption (kW)						
No subsidy (baseline)	3.46	3.36	3.18	3.41	3.41	3.69
Net metering (*status quo*)	4.23	3.73	3.46	3.75	4.19	4.65
Feed-in tariff	4.69	4.71	4.42	4.64	4.58	4.81
Investment subsidy	4.03	4.05	4.00	3.97	3.96	4.05
Lump-sum	3.48	3.35	3.32	3.44	3.41	3.64

Notes: Panel A reports the conditional adoption rate and its distribution across five income quintiles under the baseline (no subsidy) and alternative solar support policies. Panel B reports the corresponding average adopted capacity (kW) among PV adopters.

Because the counterfactuals are computed under a fixed capacity target, differences in adoption rates implies an intensive margin. Panel B of Table 4 shows that capacity per adoption is larger under production-based subsidies than under investment-based ones. There are two reasons. The first is a composition effect: under investment-based subsidies, more low-income households adopt, and they install smaller PV sizes than high-income households. The second reason is that the policy design itself encourages different optimal sizes. Specifically, a lump-sum transfer affects the adoption decision but not the adopted panel size. Therefore, the adopted capacity per adoption is very close to the baseline scenario. An investment subsidy effectively lowers the marginal installation cost and increases the optimal installed capacity. Nevertheless, the difference relative to the lump-sum transfer remains small, because for most households, the marginal revenue from wholesale electricity sales does not offset the additional investment cost. Under net metering, the marginal benefit of capacity remains at the value of retail price up to the level of a household's consumption, which is why adoption bunches at that threshold and is very different across different income quintiles with different electricity consumption. Under a feed-in tariff, since the marginal benefit is fixed at a relatively high level while marginal installation costs decline with economies of scale, adoption reaches a corner solution constrained by the physical limit of rooftop surface area.

6.2 Social cost and fiscal budget

I now discuss the cost implications of the intensive margin and change of adoption distribution of different subsidies. See Figure 4 for the summary.

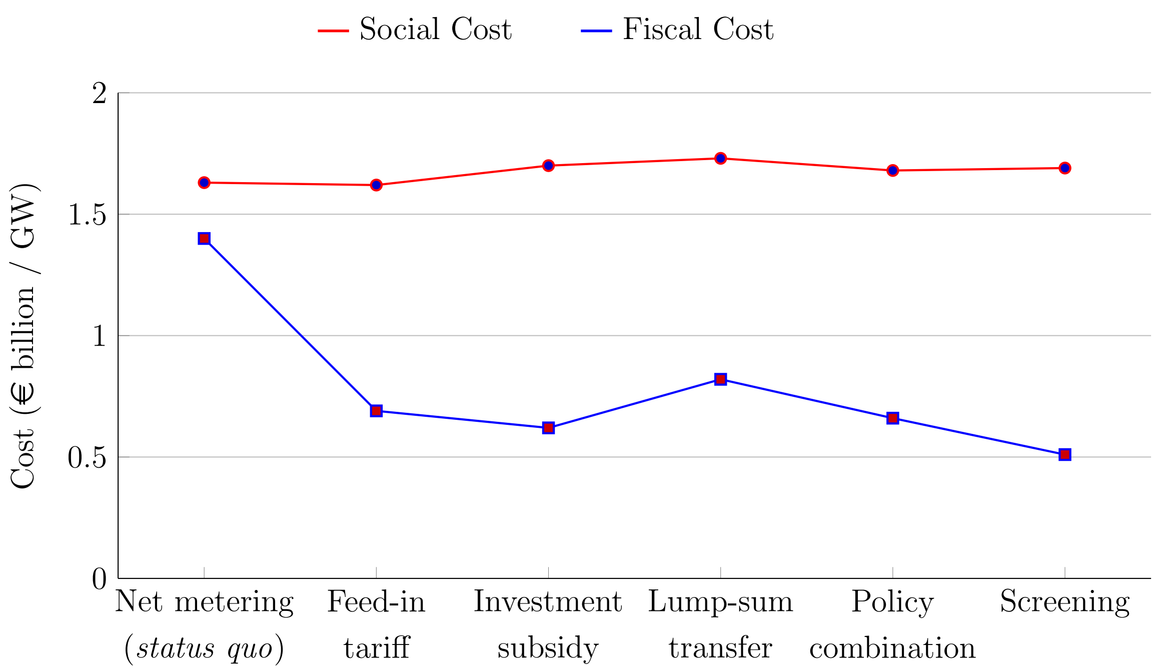


Figure 4: Optimal Policy

Note: This figure compares the social cost and fiscal budget across four single policies, policy combination, and policy screening per gigawatt (GW) of PV.

Social cost The social cost of PV adoption consists of two costs. The first is the PV installation cost, which is the private cost incurred by PV adopters. The second is the grid investment cost, which is the additional investments required to upgrade the grid to accommodate electricity feed-in that exceeds the grid's initial capacity before expansion of residential PV. This is a public cost, as grid investments are spread across all households regardless of their adoption status. The intensive margins have two countervailing cost effects.

First, although the total installed capacity is the same, the average adopted size per adoption differs across subsidies. Because of economies of scale, the average installation cost decreases as the average adopted size increases. As a result, the installation cost is highest under the lump-sum transfer (€1.53/W) and lowest under the feed-in tariff (€1.40/W).

Second, if households' consumption behavior is independent of solar adoption and the subsidy instrument implemented, a larger PV lowers the counterfactual self-consumption rate, and more electricity is fed back into the grid, thereby increasing the grid burden and raising grid investment costs. As a proxy, I use feed-in volume to calculate the grid cost difference between different policy instruments. According to Masciandaro et al. (2025), 1 Watt of residential solar under net metering requires a grid investment of €0.22. Lump-sum subsidies reduce this cost by 9.8%, whereas feed-in tariff increases it by 0.98%.[24]

Because installation and grid costs correlate with PV size in opposite ways, the social cost difference across policies is not very large. However, because economies of scale outweigh the grid investment costs of installing larger PV, the feed-in tariff incurs the lowest total social cost.[25]

Fiscal budget At first glance, it may seem surprising that the feed-in tariff, which reimburses the future production, is less expensive than a lump-sum transfer, even though households heavily discount future benefits. This happens because of the intensive margin. Consider the lump-sum transfer in which the subsidy level does not affect the PV size per adoption. Hence, the only way to achieve the capacity target is to raise the adoption rate. As a result, the government must increase the subsidy level not only for marginal adopters but also for infra-marginal ones, which is costly. This subsidy cost would be lower for the proportional investment subsidy, which can induce households to install a larger PV size when increasing the subsidy level.

In contrast, the feed-in tariff, which has the strongest intensive margin effect, allows the government to reach the same capacity target with a lower subsidy level and a lower adoption rate. Consequently, even when households heavily discount future benefits, the investment subsidy only saves 10% compared to the feed-in tariff. This difference is much smaller than that reported by De Groote and Verboven (2019), who do not consider the intensive margin and find that the investment subsidy saves

[24]If any, demand shifting is more likely to happen when investment-based subsidy is used, and hence the effect is even stronger.

[25]This result is sensitive to the grid cost. If the grid cost of €0.22 per Watt is underestimated, the social cost difference would converge. On the other hand, if the cost is overestimated, the social cost gap will widen.

51% of the fiscal cost of the feed-in tariff.

Policy combination So far, I have only considered a single subsidy instrument, while policy combinations are common in practice. I find that this yields a convex combination of single policies. For instance, Figure 4 illustrates a combination of investment subsidy and a feed-in tariff that minimizes the fiscal cost.[26] Under this constraint, the fiscal cost is reduced by approximately €30 million per gigawatt compared with the feed-in tariff, but it is more expensive than the investment subsidy.

Screening Now, I propose a more effective policy by leveraging the heterogeneity of households. Specifically, because the price coefficient ratio $\beta^C_{q_i}/\beta^R_{q_i}$ monotonically decreases with household income quintiles, it is possible to design a production-based subsidy and an investment-based subsidy such that there is a cut of income quintile q, and the income quintile lower than the threshold $\underline{q}$ chooses investment-based subsidy and those higher than the threshold $\bar{q}$ chooses production-based subsidy. The incentive-compatible constraint holds as long as the subsidy satisfies:

$$\frac{\beta^C_{\bar{q}}}{\beta^R_{\bar{q}}} < \frac{\mathcal{R}^{production-based}}{\mathcal{R}^{investment-based}} < \frac{\beta^C_{\underline{q}}}{\beta^R_{\underline{q}}} \tag{20}$$

Compared with the investment subsidy, the screening mechanism that combines the investment subsidy and the feed-in tariff can save fiscal expenditure by 18%, corresponding to a saving of €110 million per gigawatt. This improvement arises because households select the "best" policy that maximizes their utility, which leads to a lower subsidy level than under a single policy. The social cost improves by €10 million per gigawatt. Compared with the feed-in tariff, the social cost increases by €70 million per gigawatt because the average installation cost rises when low-income households reduce their PV size, whereas the fiscal cost decreases by €170 million per gigawatt.[27]

[26] I do not allow a corner solution where only the investment subsidy is used. If the feed-in tariff is allowed to be negative, so that a feed-in cost is charged, the fiscal cost can be further reduced. Such a feed-in cost may encourage self-consumption, but it may also discourage solar electricity production. I leave this as an interesting question for future research.

[27] Another advantage of screening is that it does not need to explicitly specify income thresholds or other conditions and avoids the administrative costs of eligibility verification. Moreover, screening helps alleviate the substantial upfront financial burden associated with investment subsidy while preserving its fiscal advantages.

6.3 Welfare distribution

To study the welfare effects of residential solar policy instruments, I compare the equilibrium outcomes under the four different policies with a baseline equilibrium in which subsidies are not available. I compute the equivalent variation (EV_i^{gs}), expressed in euros, which represents the monetary amount required to make a household indifferent between the subsidy and the no-subsidy scenario. In addition, I calculate the total subsidy expenditure borne by the government and assume that this expenditure is financed by all households, which I refer to as the "subsidy tax". The subsidy also incentivizes more solar adoption and hence higher grid costs, which are spread evenly across all households. Finally, I added back the environmental benefits.[28] The net welfare effect, dW_i^{gs} for household i is then given by:

$$dW_i^{gs} = EV_i^{gs} - \text{Subsidy tax}_i^{gs} - \Delta\text{Grid cost}_i + \Delta\text{Environmental benefits}_i, \quad (21)$$

and $EV_i^{gs} = \frac{1}{\bar{\beta}_i}\{\log[1 + exp(I_{igt}^{gs})] - \log[1 + exp(I_{igt}^{base})]\}$, where $\bar{\beta}_i$ is the average price sensitivity of household i.

Note that the estimation sample excludes tenants, as it is difficult to justify their decision on solar PV adoption. Therefore, tenant responses to residential solar subsidies remain unclear. In the welfare analysis, I assume that tenants cannot adopt solar panels, implying a zero adoption rate. Nonetheless, all households, regardless of home ownership, contribute to financing subsidies and grid costs and benefit from pollution reduction. To assess the robustness of this assumption, Table 6 in Appendix D reports results under the alternative assumption that tenants adopt in the same way as homeowners. Moreover, income quintiles are defined using the raw data, in which the number of households in each quintile is equal, whereas in the cleaned sample, the quintile sizes differ.[29] I assume that households in each quintile in the estimation sample and those that installed PV before 2020 are similar. Under this assumption, I rescale the number of households in each quintile to make all quintiles equal in size, and then report the welfare change for each quintile in Table 5.

I consider three cases. First, a lump-sum tax that is the same for all households. Second, a surcharge on each kilowatt-hour of net electricity consumption, which acts

[28]I refer to Sexton et al. (2021) that the average discounted environmental benefits are \$2317.5 per kW, which is equal to €2012.05 per kW, using an exchange rate of \$1 = €0.868. The interest rate is 3%.

[29]The number of households from the first to fifth quintiles is 486204, 586248, 581391, 480545, 371518, respectively. The number of households in the first quintile is low because many low-income households share the same apartment, and their energy consumption data cannot be identified and hence dropped. The number of households in the fourth and fifth quintiles is low because many wealthy households installed solar PV before 2020.

as a marginal tax so that households consuming more electricity pay more.[30] Third, financing the subsidy through higher income taxes, with the revenue raised as part of the general budget rather than an energy-specific charge.

Panel A gives the net welfare change when a lump-sum tax is charged. Note that the analysis is conducted in partial equilibrium, as it does not account for other markets, especially the effect on the labor market.[31] In addition, the effect of residential solar capacity on the wholesale electricity prices are not included.[32] I observe a negative number for net metering because of the high fiscal expenses, and households heavily discount future benefits. The screening generates the highest welfare gain due to its low fiscal costs. As expected, when subsidies are financed equally across all households, the PV subsidy is highly regressive, as high-income households receive a larger share of the benefits. This regressivity is independent of which subsidy is used.

When the subsidy is financed through a surcharge, higher-income households bear a larger share of the cost than in the case when a lump-sum tax is used. This is because high-income households consume more electricity: in our sample, households in the first quintile consume, on average, 1742 kWh per year, while those in the fifth income quintile consume 4119 kWh. At the early stage of residential solar adoption, the surcharge makes the net effect of the policy much less regressive than when the subsidy is financed through a lump-sum tax, as richer households contribute more to financing residential solar adoption. [33]

[30] Net electricity consumption is defined differently across subsidy instruments. Under net metering, it is measured as the difference between grid consumption and electricity fed into the grid. Under the other three policies, net consumption is equal to grid consumption because electricity is purchased and sold separately.

[31] The labor market effects of energy policy are an interesting and important topic, but they are beyond the scope of this paper.

[32] Masciandaro et al. (2025) estimate the merit order effect of solar PV and find it is very small. Hence, I do not expect it to overturn the main result.

[33] Although high-income households still receive more subsidies than they pay in taxes, they invest a lot in solar panel installations.

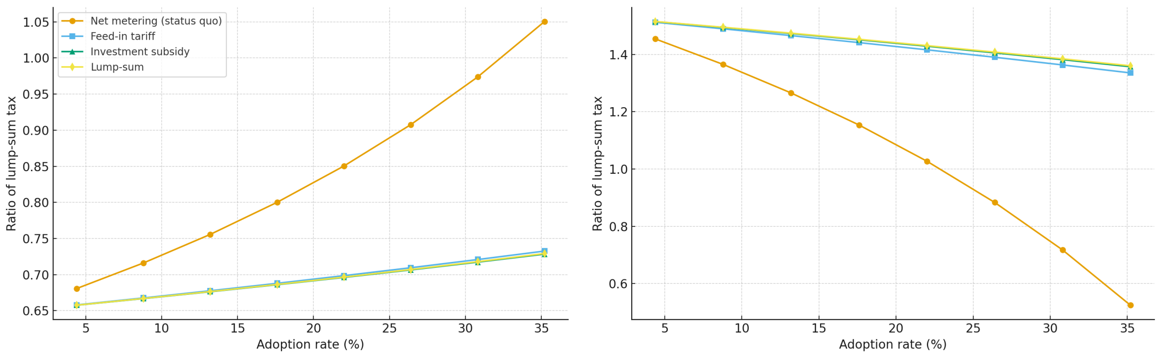
Table 5: Welfare Change of Alternative Policies (€, per HH/year)

	< 20%	20–40%	40–60%	60–80%	> 80%
Panel A: welfare change with lump-sum tax					
Net metering (*status quo*)	-74.26	-65.60	-32.70	+23.24	+88.24
Feed-in tariff	+52.99	+61.95	+87.75	+120.42	+152.23
Investment subsidy	+65.35	+82.96	+119.96	+158.67	+195.15
Lump-sum	+34.70	+61.92	+110.55	+156.72	+195.55
Screening	+85.44	+96.72	+124.14	+156.42	+188.32
Panel B: welfare change with surcharge					
Net metering (*status quo*)	+15.48	-12.67	-20.19	-21.22	-13.11
Feed-in tariff	+99.45	+90.59	+94.98	+96.57	+95.98
Investment subsidy	+106.07	+108.05	+126.27	+137.68	+145.67
Lump-sum	+88.34	+94.96	+118.81	+128.95	+130.11
Screening	+120.73	+118.16	+128.97	+137.16	+143.99
Panel C: welfare change with income tax					
Net metering (*status quo*)	+100.20	+61.39	+23.52	-24.31	-221.88
Feed-in tariff	+137.25	+123.28	+114.90	+97.45	+2.45
Investment subsidy	+139.42	+136.87	+143.83	+138.48	+63.48
Lump-sum	+132.59	+133.16	+142.10	+130.04	+21.55
Screening	+150.45	+144.04	+145.09	+138.70	+72.77

Notes: This table gives the welfare changes relative to the baseline across five income quintiles of alternative solar support policies and alternative ways to finance subsidy.

However, this result does not necessarily hold over time. Our data begins with a sample where adoption is zero. As adoption increases and the adoption rates of low- and high-income households rise at the constant pace in Table 4, the surcharge under the net metering policy becomes increasingly regressive. Specifically, once the overall adoption rate reaches around 26%, the first-quintile-income households end up paying a larger cost than households in the fifth quintile, and when the adoption rate reaches 32.3%, the households in the first quintile contribute more than in the case when an identical lump-sum tax is charged. While this pattern is less obvious under other policies. See figure 5.

Finally, the regressive effect can be largely mitigated if the subsidy is financed through income taxation. In this case, high-income households contribute more subsidies than they receive, while paying for their own installation costs, making them the main contributors to residential solar adoption.



(a) First income quintile

(b) Fifth income quintile

Figure 5: Surcharge Contribution of Households

Notes: This figure plots the change in the surcharge, expressed as a ratio of the lump-sum tax, against the total adoption rate. The subfigures show the change for households in the first income quintile and those in the fifth income quintile.

7 Robustness

This section examines the robustness of the main findings across alternative model specifications.

7.1 Exogenous capacity choice

I first examine the robustness of the results with respect to model specification. In the benchmark, I estimate a nested logit model with an outside option and capacity divided into five categories. As an alternative, I estimate a standard logit model. In this specification, households are allowed to install capacity in increments of 400W, but the adopted capacity is exogenously determined.[34] To assign this exogenous capacity, I regress the observed installed capacity of PV adopters on households' one-year lagged electricity consumption, house surface area, and household size. Hence,

$$K_{it} = \alpha_0 + \alpha_1 \, D_{i,t-1} + \alpha_2 \, \mathrm{Surf}_i + \alpha_3 \, \mathrm{HHsize}_i + \varepsilon_{it}. \tag{22}$$

I then use the predicted value $\hat{K}_{it}$ as the capacity that household i would install in the estimation sample. Based on this, I specify a logit model in which the individual utility of adoption is given by:

$$u_{i1t} = \delta_{i1t} + \epsilon_{i1t}. \tag{23}$$

[34] 400W is the standard size of a solar panel. I also tried other sizes, such as 100W and 240W, which give similar results.

with the condition value

$$\delta_{i1t} = \beta^R_{q_i}\mathcal{R}^{sq}_{i1t} - \beta^C_{q_i}C_{i1t} + +\mathbf{\Phi}\mathbf{w}_i + \boldsymbol{x_{i1t}\gamma}. \tag{24}$$

The cost and future revenue are calculated similarly to those in the nested logit model, using the predicted installed capacity. Control variables include year and alternative-location fixed effects.[35] The conditional value of non-adoption remains normalized to zero. Household characteristics are unchanged, but the parameter matrix is now independent of the capacity alternative. The full sample is used. The estimation results are reported in Table 7, which are similar to those in the main specification in Table 3.

7.2 Different discount factor

In the benchmark model, I use a discount factor of 0.85. One may question whether the results are sensitive to this assumption. To address this concern, I estimate the model using alternative discount factors of 0.9, 0.95, 0.97 and 1. The results in Table 8 indicate that the main findings remain robust.

7.3 Heterogeneous discount factors

The estimation results in Table 3 give different price coefficients conditional on all households are assigned the same discount factor, and this finding is robust across various model specifications. One explanation for this result is that poorer households have more financial constraints, so that they are more sensitive to investment costs.

However, households may discount future benefits at different rates. Specifically, lower-income households may have a smaller discount factor, reflecting that they are more short-sighted and/or more risk-averse. Figure 6 draws the ratio of cost sensitivity to future income sensitivity for discount factors ranging from 0.8 to 1.[36] It shows that the same ratio across income quintiles could be achieved when households have different discount factors. For instance, when the ratio equals one, the implied discount factors for the first to fifth quintiles are 0.87, 0.87, 0.9, 0.92, and 0.94, respectively.[37] With such heterogeneous discount factors, the observed differences in price coefficients would vanish. The policy implication between reimbursing solar PV

[35] In this logit specification, capacity can still be categorized. The difference is that households no longer choose among capacity categories as in the nested model.

[36] This is only a rough illustration of how the discount factor would change the coefficient ratio. It does not mean that the discount factor is computed by a simple division.

[37] This implies a discount factor difference of 0.07 between the first and fifth quintiles, which is substantial.

installation costs or awarding future production remains, but now the variation arises from differences in intertemporal preferences. In this case, the same amount of future income provides less incentive for lower-income households, as they discount future benefits more.

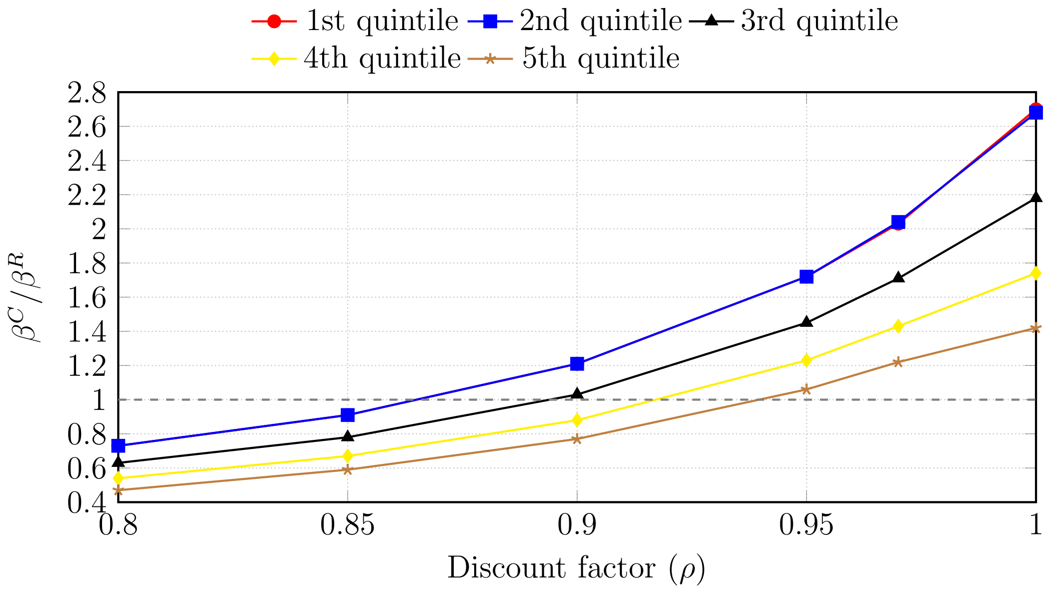


Figure 6: Price Sensitivity Ratio

Note: This figure shows the ratio between cost sensitivity and future revenue sensitivity in absolute terms, β^C/β^R, across discount factors (ρ) from 0.8 to 1, for five income quintiles.

8 Conclusion

This paper studies the cost and welfare implications of various residential PV subsidies using Dutch administrative data and a nested discrete choice model that allows households to have heterogeneous preferences and endogenous PV capacity choices.

I find that subsidies affect the intensive margin of PV adoption, which provides two new cost implications. First, the PV size installed by a household affects the installation cost through economies of scale and affects the level of self-consumption. Second, the intensive margin determines how effectively each policy achieves the total PV capacity target. In particular, investment-based policies provide weak incentives for larger PV sizes. Hence, meeting a capacity target requires increasing adoption through higher subsidy levels. This raises fiscal costs and largely offsets the advantage of investment-based policies over production-based ones, whose future benefits are heavily discounted by households.

I then propose a novel policy screening design. Instead of a single or a combination of policy instruments that uniformly apply to all people, screening allows households to select from a menu of policies by leveraging heterogeneous preferences. Such self-selection would minimize the fiscal costs while also achieving the largest welfare gain.

Regarding the welfare distribution, since low-income households are more cost-sensitive and face liquidity constraints, shifting to an investment-based subsidy would incentivize more adoption among low-income households than a production-based subsidy. This finding is consistent with Zwick and Mahon (2017), showing small firms are more incentivized by immediate tax credit, and this suggests a broader application for policy design in other sectors. However, the change of PV subsidy itself cannot solve the regressive effects, as high-income households always adopt more and receive more subsidies. Then, I show that the way subsidies are financed plays a crucial role. Specifically, financing through surcharges is much less regressive as long as the net metering policy is avoided. And income taxation is progressive.

For future research, several topics can be pursued contingent on data availability. First, the analysis is conducted in a partial equilibrium framework, ignoring labor market effects and the change of the electricity wholesale market with solar energy integration. Second, with limited credit data, I cannot disentangle the sources of households' heterogeneous responses to subsidies. Third, data limitations prevent me from capturing heterogeneity in rooftop conditions, installation costs across companies and locations, and variation in electricity retail contracts, all of which may influence adoption. Fourth, the expansion of electric vehicles and heat pumps would change the long-term electricity consumption and affect the households' adoption decision. Finally, the household choice would be better understood if non-monetary behavioral factors, such as inattention, misperceived payoffs, salience, or risk preferences, can be quantified and incorporated into the analysis.

References

Aldy, J. E., Gerarden, T. D., & Sweeney, R. L. (2023). Investment versus Output Subsidies: Implications of Alternative Incentives for Wind Energy. *Journal of the Association of Environmental and Resource Economists*, *10*(4), 981–1018. https://doi.org/10.1086/723142

Aydın, E., Brounen, D., & Ergün, A. (2023). The rebound effect of solar panel adoption: Evidence from Dutch households. *Energy Economics*, *120*, 106645. https://doi.org/10.1016/j.eneco.2023.106645

Bollinger, B., Gillingham, K., & Kirkpatrick, A. J. (2025, January). *Valuing Solar Subsidies* (tech. rep. No. w33368). National Bureau of Economic Research. https://doi.org/10.3386/w33368

Böning, J., Bruninx, K., Ovaere, M., Pepermans, G., & Delarue, E. (2025). The effectiveness of future financial benefits on PV adoption — Evidence from Belgium. *Energy Economics*, *142*, 108141. https://doi.org/10.1016/j.eneco.2024.108141

Burr, C. (2016). Subsidies and Investments in the Solar Power Market.

Cardell, N. S. (1997). Variance Components Structures for the Extreme-Value and Logistic Distributions with Application to Models of Heterogeneity. *Econometric Theory*, *13*(2), 185–213. https://doi.org/10.1017/S0266466600005727

Comello, S., & Reichelstein, S. (2017). Cost competitiveness of residential solar PV: The impact of net metering restrictions. *Renewable and Sustainable Energy Reviews*, *75*, 46–57. https://doi.org/10.1016/j.rser.2016.10.050

Crago, C. L., & Chernyakhovskiy, I. (2017). Are policy incentives for solar power effective? Evidence from residential installations in the Northeast. *Journal of Environmental Economics and Management*, *81*, 132–151. https://doi.org/10.1016/j.jeem.2016.09.008

Dauwalter, T. E., & Harris, R. I. (2023). Distributional Benefits of Rooftop Solar Capacity. *Journal of the Association of Environmental and Resource Economists*, *10*(2), 487–523. https://doi.org/10.1086/721604

De Groote, O., & Verboven, F. (2019). Subsidies and Time Discounting in New Technology Adoption: Evidence from Solar Photovoltaic Systems. *American Economic Review*, *109*(6), 2137–2172. https://doi.org/10.1257/aer.20161343

Eid, C., Reneses Guillén, J., Frías Marín, P., & Hakvoort, R. (2014). The economic effect of electricity net-metering with solar PV: Consequences for network cost recovery, cross subsidies and policy objectives. *Energy Policy*, *75*, 244–254. https://doi.org/10.1016/j.enpol.2014.09.011

Feger, F., Pavanini, N., & Radulescu, D. (2022). Welfare and Redistribution in Residential Electricity Markets with Solar Power. *The Review of Economic Studies*, *89*(6), 3267–3302. https://doi.org/10.1093/restud/rdac005

Gillingham, K., & Tsvetanov, T. (2019). Hurdles and steps: Estimating demand for solar photovoltaics. *Quantitative Economics*, *10*(1), 275–310. https://doi.org/10.3982/QE919

Holland, S. P., Mansur, E. T., Muller, N. Z., & Yates, A. J. (2019). Distributional Effects of Air Pollution from Electric Vehicle Adoption. *Journal of the As-*

sociation of Environmental and Resource Economists, *6*(S1), S65–S94. https://doi.org/10.1086/701188
Hughes, J. E., & Podolefsky, M. (2015). Getting Green with Solar Subsidies: Evidence from the California Solar Initiative. *Journal of the Association of Environmental and Resource Economists*, *2*(2), 235–275. https://doi.org/10.1086/681131
Ito, K., Ida, T., & Tanaka, M. (2023). Selection on Welfare Gains: Experimental Evidence from Electricity Plan Choice. *American Economic Review*, *113*(11), 2937–2973. https://doi.org/10.1257/aer.20210150
Känzig, D. R. (2023). The Unequal Economic Consequences of Carbon Pricing. *National Bureau of Economic Research.*, *No. w31221.*
Londo, M., Matton, R., Usmani, O., van Klaveren, M., Tigchelaar, C., & Brunsting, S. (2020). Alternatives for current net metering policy for solar PV in the Netherlands: A comparison of impacts on business case and purchasing behaviour of private homeowners, and on governmental costs. *Renewable Energy*, *147*, 903–915. https://doi.org/10.1016/j.renene.2019.09.062
Masciandaro, C., Mulder, M., & Kesina, M. (2025). Distributional effects of the Dutch net-metering scheme for residential solar panels. *Energy Economics*, *151*, 108891. https://doi.org/10.1016/j.eneco.2025.108891
McFadden, D. (1977). Modelling the Choice of Residential Location.
Sexton, S., Kirkpatrick, A. J., Harris, R. I., & Muller, N. Z. (2021). Heterogeneous Solar Capacity Benefits, Appropriability, and the Costs of Suboptimal Siting. *Journal of the Association of Environmental and Resource Economists*, *8*(6), 1209–1244. https://doi.org/10.1086/714970
Wolak, F. (2018, September). *The Evidence from California on the Economic Impact of Inefficient Distribution Network Pricing* (tech. rep. No. w25087). National Bureau of Economic Research. Cambridge, MA. https://doi.org/10.3386/w25087
Zwick, E., & Mahon, J. (2017). Tax Policy and Heterogeneous Investment Behavior. *American Economic Review*, *107*(1), 217–248. https://doi.org/10.1257/aer.20140855

Appendix

A Dutch electricity markets

Electricity activities include production, wholesaling, transmission, distribution, and retailing. Transmission and distribution are not liberalized; therefore, electricity markets typically refer to wholesale and retail markets. Electricity is traded on many wholesale markets, including over-the-counter markets, long-term forward and futures markets, day-ahead markets, intraday markets, and balancing markets. There are also ancillary markets that operate as backups to maintain grid stability and efficiency. The *day-ahead wholesale market* is one of the most important markets for electricity trading in the Netherlands. The day-ahead wholesale market works as follows: on day $d-1$ before noon, electricity sellers and buyers bid prices on the volume of electricity they are willing to buy or sell for each hour h of the day d. After the gate closes, the market operator matches the bids in a merit order, meaning that dispatch starts from the cheapest fuel. Then, the market operator determines the market-clearing price for each hour by the marginal cost of the most expensive supply bid needed to meet the demand, and all the selected bids are settled at the same price. Without further explanation, the wholesale electricity price in the rest of the paper refers to the price set in the day-ahead wholesale market. As renewable energy has zero marginal cost, it shifts the supply curve to the right (from blue to red) and lowers the market-clearing price. See Figure 7.

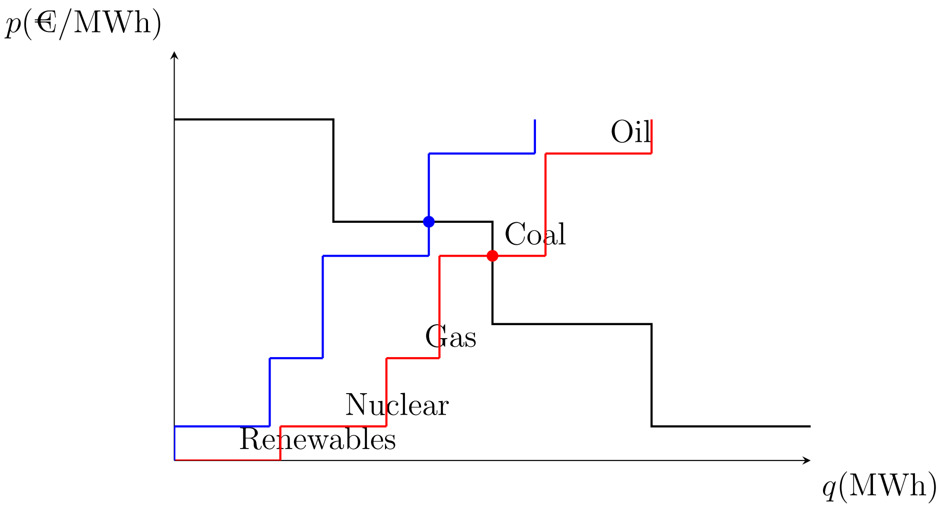


Figure 7: Day-ahead Market Auction

Notes: This figure shows how the day-ahead wholesale electricity price is determined in a blind auction. The operator collects upward supply bids and downward demand bids. The market-clearing price and quantity are determined by the intersection of the supply and demand curves. The figure is illustrative and does not represent the actual market structure in the Netherlands.

Most energy companies are integrated, meaning they engage in both electricity production and retailing. However, in this paper, I assume that wholesale and retail activities are independent. Hence, Retail activity refers to energy companies purchasing electricity from the wholesale markets and reselling it to households. Most households are not directly exposed to wholesale electricity prices. Instead, energy companies set energy prices that are typically calculated as a weighted average of wholesale prices.

The retail price, however, includes not only the energy price but also a fixed grid cost charged by grid operators and taxes paid to the government. Government taxes are levied on the volume of net electricity consumption (hence the consumption from the grid minus the volume fed back into the grid), including energy tax and sustainable energy surcharge (ODE-heffing).[38] As energy is a basic need, each household receives a lump-sum tax refund. This refund is the same for every household, independent of their electricity consumption, income, house type, and solar PV adoption status. Finally, the retail price is subject to a value-added tax (henceforth, VAT) of 21 percent.[39] As fixed costs are independent of the amount of electricity consumption and net metering, from now on, VAT included retail price R only refers to the volumetric retail price (hence energy price r, plus taxes) and excludes fixed delivery and grid costs, and tax refund. Denote τ_e as the sum of energy tax and sustainable energy surcharge per kWh, and τ_v as VAT. The per-unit tax on electricity consumption is defined as

$$\tau = r * \tau_v + \tau_e(1 + \tau_v) \tag{25}$$

The retail price per kWh is

$$R = r + \tau \tag{26}$$

Before 2022, 72% of the retail price was taxes and levies. Only 28% was for the energy price. In 2022 and 2023, temporarily low taxes and high electricity prices reduced the ratio of taxes to 42%. See Figure 8.

In the Netherlands, the electricity bill is issued annually, while the settled price is determined based on the type of retail contracts. There are three types of retail contracts: fixed, variable,[40] and dynamic. The fixed contract charges a fixed retail price per kWh for a certain period between one to three years, while the variable contract has a variable price per kWh that changes twice to four times a year and

[38] ODE tax was collected to subsidize renewable investment from 2013 and has been canceled as a separate tax and included in the energy tax since 2023.

[39] VAT rate was 19% until October 2012 and temporarily reduced to 9% between July and December of 2022 because of the energy crisis.

[40] Model contract is a special variable contract that each energy company is obliged to offer, and the conditions of the model contract are the same across all suppliers, so consumers can easily compare and switch between different suppliers. However, the energy prices could differ. Besides model contracts, energy companies can also choose to offer other variable contracts that have different conditions and rates.

can be canceled monthly.[41] The dynamic contract is new in the Netherlands, starting from the second half of 2022. It provides a price that changes daily or hourly based on the day-ahead wholesale electricity prices, plus a purchase fee per kWh. By the end of 2023, 27 out of 50 suppliers provide dynamic contracts. In total, 50% of households choose fixed contracts, 47% variable contracts, and only 3% dynamic contracts.

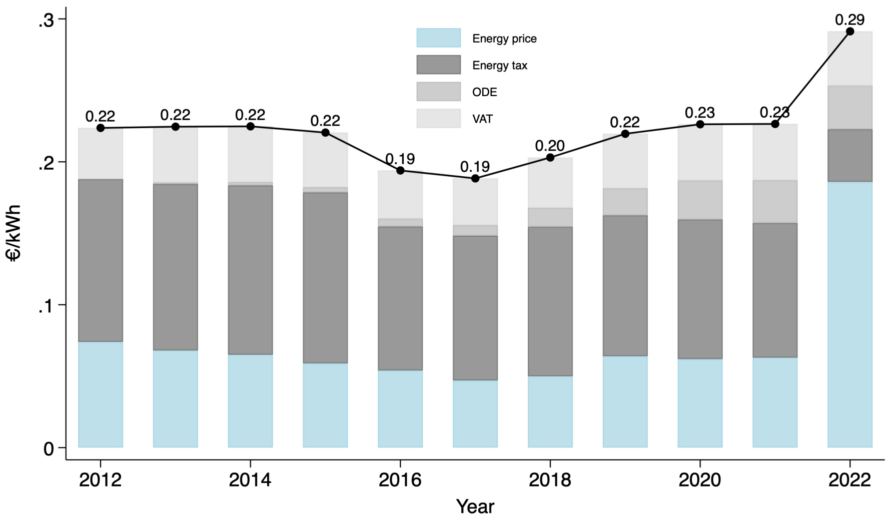
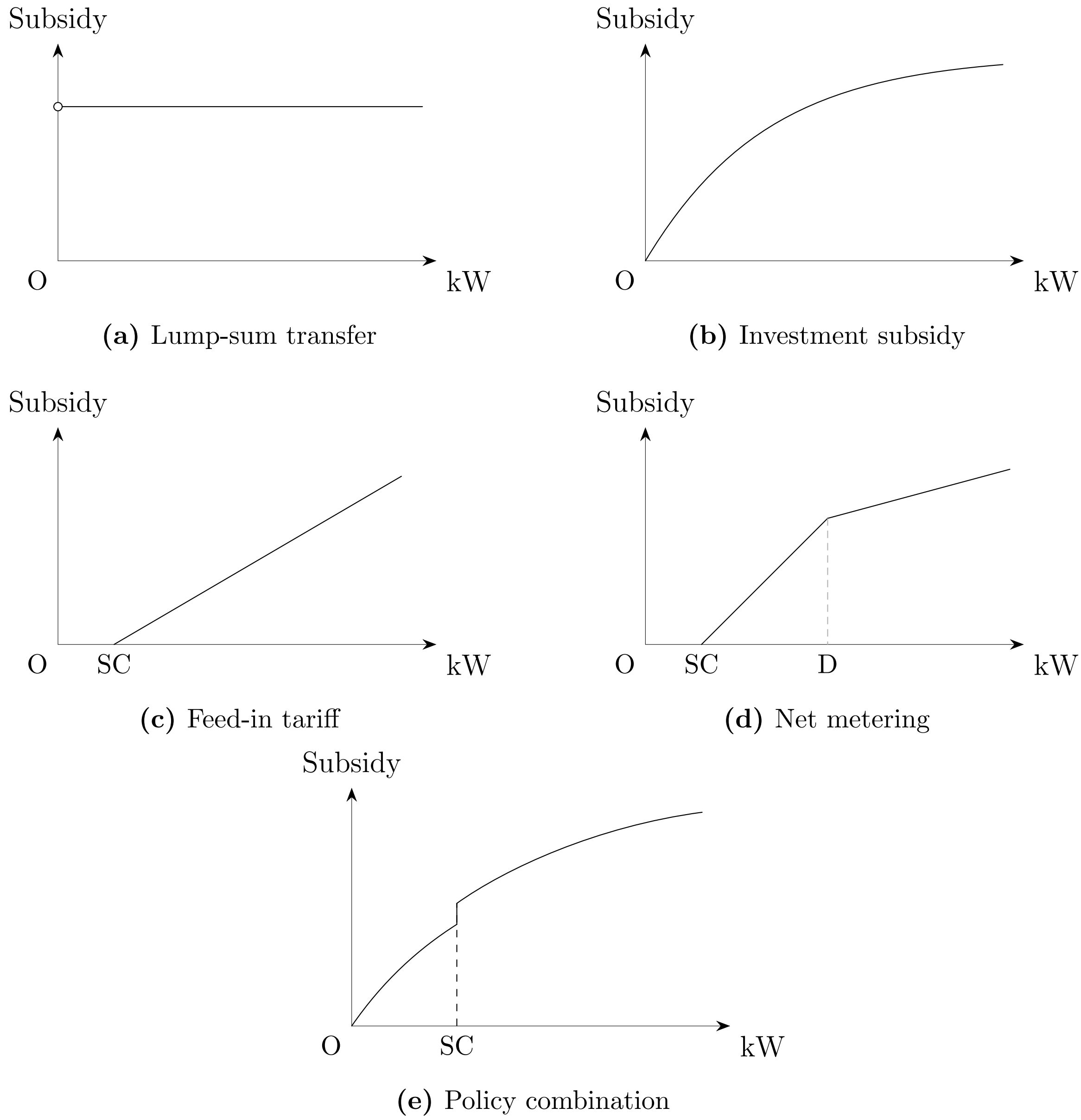


Figure 8: Retail Electricity Price (€/kWh)

Notes: This figure shows the average variable retail price and its breakdown in the Netherlands from 2012 to 2022.

[41]Variable rates were adjusted on Jan 1 and July 1 until the energy crisis, when suppliers could adjust more frequently based on market conditions.

B Subsidy structure



(a) Lump-sum transfer
(b) Investment subsidy
(c) Feed-in tariff
(d) Net metering
(e) Policy combination

Figure 9: Subsidy Structure

Notes: The figure illustrates the subsidy as a function of capacity: (a) Under a lump-sum transfer, the subsidy is independent of capacity. (b) Under an investment subsidy, the curve is concave because the marginal investment cost decreases with capacity. (c) Under a feed-in tariff, the curve starts with self-consumption and after that, increases linearly with capacity. (d) Under net metering, the curve increases steeply up to the level of household demand, beyond which additional capacity yields much smaller returns. (e) A combination of feed-in tariff and investment subsidy. SC refers to self-consumption. D refers to electricity consumption.

C Solar Efficiency and Self-consumption

This section describes the calibration method for solar efficiency and self-consumption rate, which are used for structural estimation and policy discussions.

Self-consumption Define $Y_{d,h}$ as the solar production on hour h of day d. Define Y as the total annual solar production, $Y = \sum_d \sum_h Y_{d,h}$. $\mathbf{y} = \{\frac{Y_{d,h}}{Y}\}$ is the production profile. $X_{d,h}$ and $Z_{d,h}$ are the electricity consumption and feed-in with the same notation. the profile $\mathbf{x}$ and $\mathbf{z}$ are defined similarly. Self-consumption rate is

$$\alpha = \frac{Y - Z}{Y} \tag{27}$$

αY is the total self-consumption, so the total consumption of PV adopters equals grid consumption plus self-consumption, $X_1 + \alpha Y$.

There are no official data on the self-consumption rate, which is estimated to be between 20-40% in the Netherlands (Londo et al., 2020). In this paper, I estimate the self-consumption rate rather than arbitrarily choosing a number for two reasons. First, the data, assumptions, and estimation methods explained later are transparent, ensuring traceable estimates. Second, the self-consumption rate is indispensable to estimate solar efficiency, which is essential to evaluate the profitability of solar systems and affect the adoption decision of households estimated in Section 3. Finally, this is useful when understanding the adoption patterns in Section 5 and welfare implications among different incentive policies in Section 6. Hence, a self-evident estimate guarantees consistency throughout the paper.

The data used for estimating the self-consumption rate are grid consumption and feed-in profiles for households with and without solar panels. Hence, $\mathbf{x_0} = \{x^0_{d,h}\}, \mathbf{x_1} = \{x^1_{d,h}\}, \mathbf{z} = \{z_{d,h}\}$. The main assumption to estimate α is that the electricity consumption profile does not change with PV adoption. In other words, there is no significant difference between the consumption profile of PV adopters and non-adopters, defined as $\mathbf{c_1} = \{c^1_{d,h}\}$ and $\mathbf{c_0} = \{c^0_{d,h}\}$, respectively. The assumption means

$$\mathbf{c_0} = \mathbf{c_1} \tag{28}$$

This is a strong assumption, but it is reasonable when net metering applies, as the surplus electricity returned to the grid can 100% offset the electricity supplied from the grid at a retail price. Hence, there is no incentive to shift demand. One concern is that consumers may adapt their consumption behavior to retail price structure: consumers shift demand to the period when the price is lower. However, the data does not support this argument, showing that consumers who are charged single, double, or night fare do not behave differently. Another concern is that consumers use more electricity when they have independent producers. I do not rule out the possibility that total consumption changes with adoption, but I restrict consumption to unaffected profiles.[42] Solar production $Y_{d,h}$ is unobservable, meaning I cannot calculate each element in consumption profiles of PV adopters $\mathbf{c_1}$. To address this issue, I use the fact that solar panels do not work at night, so consumption during those periods is only supplied from the grid.[43]

$$c^1_{d,h} = \frac{X^1_{d,h}}{X_1 + \alpha Y} = \frac{X^1_{d,h}}{X_1 + \frac{\alpha}{1-\alpha} Z} = \frac{x^1_{d,h}}{1 + \frac{\alpha}{1-\alpha} \cdot \frac{Z}{X_1}}, \tag{29}$$

Z/X_1 is the total feed-in over the total grid consumption for PV adopters, which is calculated as 0.64. For non-adopters, grid consumption is equal to total electricity consumption. Hence, $\mathbf{c_0} = \mathbf{x_0}$. The OLS regression implied by equation (28) and (29) is

$$x^0_{d,h} = \frac{x^1_{d,h}}{1 + 0.64 \cdot \alpha/(1-\alpha)} + \epsilon_{d,h}, \quad d,h \in \{d,h|z_{d,h} = 0\} \tag{30}$$

Even though α enters equation (30) in a non-linear way, $\frac{1}{1+0.64\cdot\alpha/(1-\alpha)}$ can be estimated by a simple OLS regression, and the estimated self-consumption rate $\hat{\alpha}$ is uniquely determined. The result is $\hat{\alpha} = 0.33$, suggesting that households, on average, directly use 33% of electricity produced by residential solar systems. The number aligns with the range suggested by Londo et al. (2020), hence a reasonable estimate.

Solar Efficiency CBS provides data on households' solar PV adoption status, capacity, and yearly electricity grid consumption and feed-in from 2019 to 2022. I do not observe the specific installation time. To estimate yearly solar production, I identify households installing PV in 2020 and track their electricity feed-ins in 2021. Similarly, for PV installed in 2021, I track the feed-ins in 2022. All other observations are dropped.

Let i index household and y index year. Denote K_i the solar installed capacity. By definition, $\iota = Y_{i,y}/K_i$, and $Y_{i,y} = Z_{i,y}/(1-\alpha)$. Hence,

$$\iota = \frac{Z_{i,y}}{(1-\alpha)K_i} \tag{31}$$

Hence, the OLS regression is

$$Z_{i,y} = \iota(1-\alpha)K_i + \mathcal{E}_{i,y} \tag{32}$$

[42]The violation of this assumption may overestimate the self-consumption rate as consumption is more likely to increase in the daytime when solar produces.

[43]Due to high investment costs and the net metering policy, household energy storage is uncommon in the Netherlands and is therefore not considered in this paper.

$\widehat{\iota(1-\alpha)}$ is estimated to be 0.61, indicating that one additional kWp in PV capacity would, on average, result in an additional 610 kWh feed-in to the grid per year. Using the self-consumption rate $\hat{\alpha} = 0.33$ estimated in equation (30), a simple calculation yields $\hat{\iota} = 0.91$.[44] Hence, in the Netherlands, 1kWp of PV capacity can, on average, produce 910kWh per year. Households directly consume 33% of generated electricity, and the remaining 67% is returned to the grid.

D Welfare Change

Table 6: Welfare Change: Tenants Adopt (€, per HH/year)

	< 20%	20–40%	40–60%	60–80%	> 80%
Panel A: welfare change with lump-sum tax					
Net metering (*status quo*)	-74.63	-73.30	-39.86	+18.03	+76.41
Feed-in tariff	+105.72	+107.01	+130.15	+161.66	+187.81
Investment subsidy	+159.02	+161.75	+188.25	+222.54	+250.46
Lump-sum	+147.64	+150.71	+179.63	+218.06	+244.88
Screening	+155.52	+159.06	+181.42	+211.60	+237.40
Panel B: welfare change with surcharge					
Net metering (*status quo*)	+45.74	-2.78	-23.30	-40.67	-59.96
Feed-in tariff	+169.83	+146.32	+139.85	+128.56	+109.80
Investment subsidy	+217.97	+197.85	+197.02	+191.90	+178.41
Lump-sum	+227.41	+199.46	+191.25	+176.28	+146.90
Screening	+204.44	+188.59	+187.85	+184.72	+175.61
Panel C: welfare change with income tax					
Net metering (*status quo*)	+154.96	+93.81	+34.12	-44.55	-331.68
Feed-in tariff	+221.66	+191.40	+167.51	+130.06	-18.27
Investment subsidy	+265.69	+239.39	+222.63	+193.47	+60.85
Lump-sum	+292.03	+255.81	+226.16	+178.70	-11.78
Screening	+245.28	+224.39	+210.35	+187.14	+77.84

Notes: This table gives the welfare changes relative to the baseline across five income quintiles of alternative solar support policies, assuming that tenants also adopt PV.

[44] Self-consumption rate can be estimated by data from 2020 to 2023, while the necessary data to estimate solar efficiency only span from 2021 to 2022. For consistency, I only use data in 2021 and 2022. The self-consumption rate is solved to be $\alpha = 0.35$ in 2021, and $\alpha = 0.32$ in 2022. The total feed-in is 59% of installed capacity in 2021 and 62% in 2022. The simple regression of grid-in volume over installed capacity can explain over 90% of the total variation.

E Robustness

Table 7: Robustness: Logit Specification

	(1)	(2)	(3)
β^R (€10^3)	0.417*** (0.007)	0.458*** (0.008)	0.357*** (0.009)
$-\beta^C$ (€10^3)	-0.208*** (0.008)	-0.232*** (0.008)	-0.174*** (0.009)
$\Delta\beta^R_{q_1}$		-0.194*** (0.005)	0.349*** (0.027)
$\Delta\beta^R_{q_2}$		-0.182*** (0.002)	0.324*** (0.013)
$\Delta\beta^R_{q_3}$		-0.054*** (0.001)	0.093*** (0.009)
$\Delta\beta^R_{q_4}$		-0.011*** (0.001)	-0.014* (0.008)
$-\Delta\beta^C_{q_1}$	-0.163*** (0.004)		-0.434*** (0.021)
$-\Delta\beta^C_{q_2}$	-0.149*** (0.002)		-0.394*** (0.010)
$-\Delta\beta^C_{q_3}$	-0.049*** (0.001)		-0.125*** (0.007)
$-\Delta\beta^C_{q_4}$	-0.011*** (0.001)		-0.001 (0.007)
Apartment	-2.443*** (0.015)	-2.443*** (0.015)	-2.444*** (0.015)
HH age	-0.004*** (0.0002)	-0.004*** (0.0002)	-0.004*** (0.0002)
House age	0.083*** (0.001)	0.083*** (0.001)	0.083*** (0.001)
Wealth (€10^5)	-0.011*** (0.0004)	-0.011*** (0.0004)	-0.012*** (0.0004)
Income (€10^4)	0.006*** (0.001)	0.008*** (0.001)	0.007*** (0.001)
House size ($100m^2$)	0.200*** (0.006)	0.200*** (0.006)	0.200*** (0.006)
HH size	-0.005** (0.002)	-0.005** (0.002)	-0.005** (0.002)
Year 2022	0.623*** (0.012)	0.602*** (0.012)	0.646*** (0.012)
Constant	-3.060*** (0.422)	-3.329*** (0.422)	-2.706*** (0.424)
# of Obs	4,245,808	4,245,808	4,245,808

Notes: This table reports Logit estimation results for PV adoption from 2020 to 2022. Standard errors in parentheses are clustered at the household level. Fixed effects are included in the estimation but not reported. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Table 8: Robustness: Different Discount Factor

	$\rho = 0.9$	$\rho = 0.95$	$\rho = 0.97$	$\rho = 1$
β^R (€10^3)	0.230*** (0.011)	0.157*** (0.007)	0.127*** (0.006)	0.084*** (0.003)
$-\beta^C$ (€10^3)	-0.177*** (0.011)	-0.167*** (0.011)	-0.155*** (0.011)	-0.119*** (0.011)
$\Delta\beta^R_{q_1}$	0.166*** (0.032)	0.113*** (0.019)	0.092*** (0.019)	0.057*** (0.019)
$\Delta\beta^R_{q_2}$	0.162*** (0.015)	0.110*** (0.010)	0.090*** (0.008)	0.060*** (0.003)
$\Delta\beta^R_{q_3}$	0.083*** (0.008)	0.056*** (0.005)	0.046*** (0.004)	0.031*** (0.002)
$\Delta\beta^R_{q_4}$	0.031*** (0.006)	0.020*** (0.004)	0.016*** (0.004)	0.010*** (0.004)
$-\Delta\beta^C_{q_1}$	-0.300*** (0.038)	-0.297*** (0.038)	-0.290*** (0.038)	-0.261*** (0.038)
$-\Delta\beta^C_{q_2}$	-0.296*** (0.016)	-0.292*** (0.016)	-0.287*** (0.016)	-0.266*** (0.016)
$-\Delta\beta^C_{q_3}$	-0.144*** (0.009)	-0.142*** (0.009)	-0.140*** (0.009)	-0.131*** (0.009)
$-\Delta\beta^C_{q_4}$	-0.052*** (0.007)	-0.050*** (0.007)	-0.049*** (0.007)	-0.046*** (0.007)
Apartment	-2.395*** (0.048)	-2.393*** (0.048)	-2.392*** (0.048)	-2.389*** (0.048)
HH age	-0.003*** (0.001)	-0.003*** (0.001)	-0.004*** (0.001)	-0.004*** (0.001)
House age	0.089*** (0.003)	0.089*** (0.003)	0.089*** (0.003)	0.089*** (0.003)
Wealth (€10^5)	-0.013*** (0.001)	-0.013*** (0.001)	-0.013*** (0.001)	-0.013*** (0.001)
Income (€10^4)	0.008*** (0.002)	0.008*** (0.002)	0.008*** (0.002)	0.008*** (0.002)
House size ($100m^2$)	0.061*** (0.003)	0.061*** (0.003)	0.061*** (0.003)	0.061*** (0.003)
HH size	0.007*** (0.002)	0.006*** (0.002)	0.006*** (0.002)	0.005*** (0.002)
Year 2022	0.730*** (0.020)	0.712*** (0.021)	0.696*** (0.021)	0.696*** (0.021)
σ	0.760*** (0.009)	0.752*** (0.010)	0.746*** (0.010)	0.732*** (0.010)
# of Obs	2,314,295	2,314,295	2,314,295	2,314,295

Notes: This table reports the estimation results of nest logit model for PV adoption from 2020 to 2022, considering different discount factors. Standard errors in parentheses are clustered at the household level. Constants and other fixed effects are included in the estimation but not reported. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.